Lecture 8 Practice List: Detailed Solutions

Economics of Organizations and Incentives (MPE Insper)

Solutions to the 25 exercises of `Listas/pratica-lecture-08.pdf`
(Tadelis Ch. 15–16, Gibbons Ch. 4, Muñoz-Garcia Ch. 9 and 11, MWG §13.C)

November 2026

How to read. Each item states what is asked, the method (with the book pages), the solution, a verification note and a boxed answer. In every perfect Bayesian equilibrium (PBE) the solution gives the strategies, the beliefs on the path (Bayes' rule) and off the path (chosen to deter deviations, with the range that works), and sequential rationality at every information set. Messages and replies are written as in the exercise. This unit is *estimated*: Lecture 8 has no slides yet, so the references are to the books. Tadelis prints no answers; Muñoz-Garcia prints worked answers for items 4, 8, 11, 14 and 23, and each of those items records that the printed answer agrees. The solutions of MWG 13.C.1, 13.C.2, 13.C.4 and 13.C.5 were compared with the MWG solutions manual; they agree except for 13.C.4, where the manual's formula drops the lowest type's boundary condition (item 25 explains). All numbers and equilibrium sets are asserted by `check_T1.py`, `check_T2.py` and `check_T3.py`.

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Lecture 8 — Perfect Bayesian equilibrium and signaling (estimated unit)

Item 1: Tadelis Exercise 15.1

Foundations · difficulty 2/5

Statement (verbatim from the list, item 1)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 15.1 "Equilibrium Selection", p. 314 [PDF p. 333]. Reproduced verbatim. Answer: none in the book.

> **Equilibrium Selection:** Consider the extensive-form game in Figure 15.5.
>
> a. Find all the Bayesian Nash equilibria of this game.
>
> b. Which of the Bayesian Nash equilibria are also perfect Bayesian equilibria? Why?

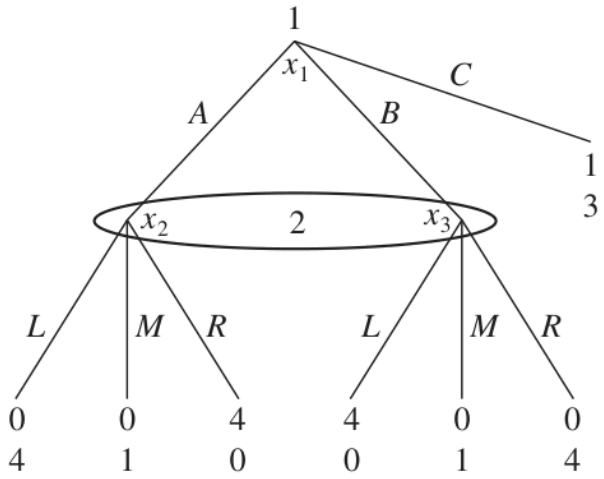


FIGURE 15.5 Exercise 15.1.

FIGURE 15.5 Exercise 15.1.

(a) What is asked. (a) All Bayesian Nash equilibria (BNE) of the tree of Figure 15.5; (b) which of them are perfect Bayesian equilibria (PBE), and why.

(b) Method. Write the normal form and find its Nash equilibria; a BNE of a tree is a Nash equilibrium of its normal form. For a PBE add a belief $\mu = \Pr(x_2)$ at player 2's information set $\{x_2, x_3\}$ and require his action to be optimal for it, on or off the path (Tadelis §15.2, Requirements 15.1–15.3, pp. 303–312; Gibbons §4.1, pp. 175–183).

(c) Solution. Payoffs (u_1, u_2); player 1 chooses A, B or C (which ends the game at $(1, 3)$), player 2 chooses L, M or R without seeing A versus B.

	L	M	R
A	0, $\underline{4}$	0, 1	$\underline{4}$, 0
B	$\underline{4}$, 0	0, 1	0, $\underline{4}$
C	1, $\underline{3}$	$\underline{1}$, $\underline{3}$	1, $\underline{3}$

(a) Pure BNE. The only cell with both payoffs underlined is (C, M): against M player 1 gets 1 from C and 0 from A or B, and player 2 is indifferent among all three replies to C.

All BNE. Let player 1 mix (a, b, c) and player 2 (l, m, r). If $a+b>0$ the information set is reached and, with $\mu = a/(a+b)$, player 2's payoffs are

$$u_2(L) = 4\mu, \qquad u_2(M) = 1, \qquad u_2(R) = 4(1-\mu).$$

M is optimal only if $4\mu \le 1$ and $4(1-\mu) \le 1$, which is impossible: whenever the information set is reached $m = 0$. Then:

- $c = 1$: player 1 gets 1; the deviations A and B pay $4r$ and $4l$, so the profile is a BNE iff $l \le \frac{1}{4}$ and $r \le \frac{1}{4}$ ($m \ge \frac{1}{2}$ is free). This family contains (C, M).
- $a, b > 0$: $m = 0$ and player 1 is indifferent between A and B iff $4r = 4l$, so $l = r = \frac{1}{2}$. Player 2 mixes L and R only if $\mu = \frac{1}{2}$, i.e. $a = b$. Both give player 1 $2 > 1$, so $c = 0$: the profile $(\frac{1}{2}A + \frac{1}{2}B,\ \frac{1}{2}L + \frac{1}{2}R)$.
- $a > 0 = b$ (or the mirror image): $\mu = 1$, player 2 plays L, and then A pays $0 < 4$ for B. No equilibrium. Mixing C with A or B would need $4r = 1$ and $4l = 1$ with $m = 0$: impossible.

(b) PBE. In the first family the information set is off the path, so μ is free, but M is a best reply for no μ: it is strictly dominated at both nodes by $\frac{1}{2}L + \frac{1}{2}R$ ($2 > 1$). Every profile of the family with $m > 0$ is therefore not sequentially rational, and $m = 0$ would force $l + r = 1 > \frac{1}{2}$. So (C, M) and the whole family are BNE but not PBE: player 2's reply M is not credible. In the second profile Bayes' rule gives $\mu = \frac{1}{2}$ and player 2 is indifferent between L and R ($2 = 2 > 1$), so mixing is sequentially rational, and player 1 is optimal ($u_1(A) = u_1(B) = 2 > 1$): a PBE.

(e) Verification. `check_T1.py` finds the pure equilibria of the matrix, enumerates every equilibrium with probabilities in multiples of $\frac{1}{12}$ and compares them with the families above, and checks over a grid of beliefs that M is never a best reply once the set is reached. Tadelis prints no answer.

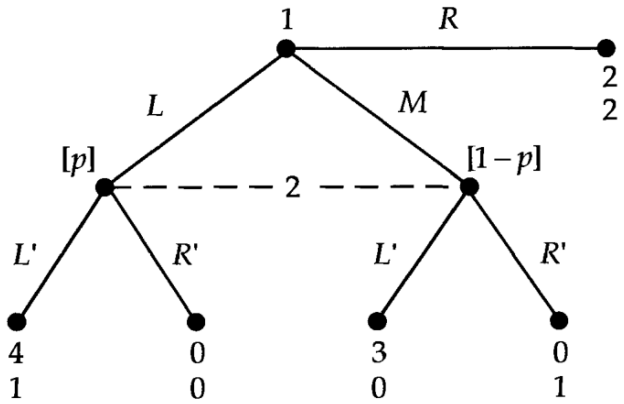
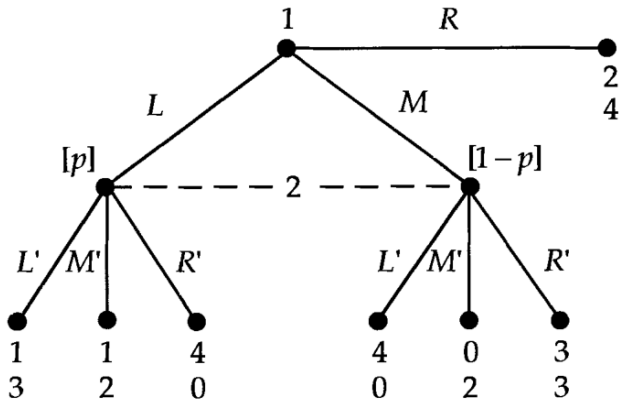
Answer. (a) BNE: C with $\Pr(L) \le \frac{1}{4}$, $\Pr(R) \le \frac{1}{4}$ (pure case (C, M)), and $(\frac{1}{2}A + \frac{1}{2}B,\ \frac{1}{2}L + \frac{1}{2}R)$. (b) Only the mixed profile, with $\mu = \frac{1}{2}$, is a PBE; (C, M) is not, because M is never optimal at player 2's information set.

Item 2: Gibbons Problem 4.1

Foundations · difficulty 3/5

Statement (verbatim from the list, item 2)

Source. Gibbons (1992), *Game Theory for Applied Economists*, §4.1 Problems, Problem 4.1, p. 245 [PDF p. 251]. Reproduced verbatim. Answer: none in the book.

> 4.1. In the following extensive-form games, derive the normal-form game and find all the pure-strategy Nash, subgame-perfect, and perfect Bayesian equilibria.
>
> **a.**
>
> 
>
>
> **b.**
>
> 
>

(a) What is asked. For each of the two trees: the normal form, and all pure-strategy Nash, subgame-perfect and perfect Bayesian equilibria.

(b) Method. Normal form, then equilibria; subgame perfection via the proper subgames (the information set of player 2 joins his two nodes, so the only subgame is the whole game); PBE by adding the belief $p = \Pr(L)$ on player 2's information set (Gibbons §4.1, pp. 175–183, Requirements 1–4; Tadelis §15.2).

(c) Solution. **Game (a).** Payoffs (u_1, u_2); R ends the game at $(2, 2)$. Player 2's payoff from L' is $p \cdot 1 + (1-p) \cdot 0 = p$ and from R' is $1 - p$: L' is optimal iff $p \geq \frac{1}{2}$.

	L'	R'
L	$\underline{4},\ \underline{1}$	$0,\ 0$
M	$3,\ 0$	$0,\ \underline{1}$
R	$2,\ \underline{2}$	$\underline{2},\ \underline{2}$

Nash: (L, L') and (R, R'); M is never a best reply. Subgame perfect: the whole game is the only subgame, so the same two. PBE: (L, L') with $p = 1$ (Bayes' rule; L' optimal since $1 \geq \frac{1}{2}$). (R, R') with any off-path belief $p \leq \frac{1}{2}$ (R' optimal iff $1 - p \geq p$); for $p > \frac{1}{2}$ player 2 would play L' and player 1 would deviate to L.

Game (b). Player 2's payoffs: L': $3p$, M': 2, R': $3(1-p)$, so M' is optimal iff $p \in [\frac{1}{3}, \frac{2}{3}]$.

	L'	M'	R'
L	$1,\ \underline{3}$	$1,\ 2$	$\underline{4},\ 0$
M	$\underline{4},\ 0$	$0,\ 2$	$3,\ \underline{3}$
R	$2,\ \underline{4}$	$\underline{2},\ \underline{4}$	$2,\ \underline{4}$

Nash: only (R, M') (against L' player 1 prefers M, against R' he prefers L). Subgame perfect: the same. PBE: (R, M') with off-path belief $p \in [\frac{1}{3}, \frac{2}{3}]$; outside that interval L' or R' is strictly better for player 2, and then player 1 deviates to M or L.

(e) Verification. `check_T1.py` builds both matrices, lists the pure equilibria and scans beliefs on a grid of step $\frac{1}{60}$ for the thresholds $\frac{1}{2}$ and $[\frac{1}{3}, \frac{2}{3}]$. Gibbons prints no answer.

Answer. (a) Nash = subgame-perfect: (L, L') and (R, R'); PBE: (L, L') with $p = 1$ and (R, R') with $p \leq \frac{1}{2}$. (b) Nash = subgame-perfect: (R, M'); PBE: (R, M') with $p \in [\frac{1}{3}, \frac{2}{3}]$.

Item 3: Tadelis Exercise 15.3

Foundations · difficulty 3/5

Statement (verbatim from the list, item 3)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 15.3 "Trust Thy Neighbor", pp. 315–316 [PDF pp. 334–335]. Reproduced verbatim. Answer: none in the book.

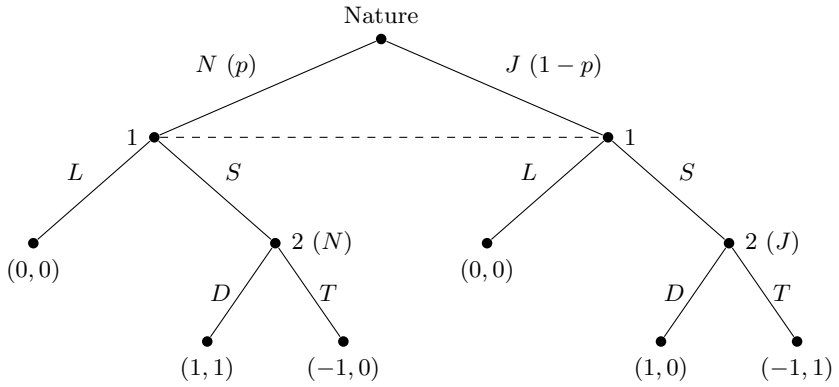
> **Trust Thy Neighbor:** A student, player 1, has to hand in a problem set at the other end of (a large) campus but needs to rush into an exam. She has two options. She can deliver the problem set after the exam (call this option L) and incur a late penalty. Alternatively she can give the problem set to player 2, a random student who happens to be next to player 1 (call this option S). Player 2 can then either deliver the problem set on time (call this option D) or throw it away in the nearest trash can (call this option T). For player 1 the payoff is 1 if the problem set is delivered on time, -1 if it is thrown away, and 0 if it is delivered late. The payoffs for player 2 are x if he delivers and y if he throws it away.
>
> a. Draw the game tree that represents this game. Using equilibrium analysis, what conditions on x and y would justify player 1 trusting player 2 to deliver the problem set?
>
> b. Now assume that some proportion p of the population of students are "nice guys" (N), for which $x = 1$ and $y = 0$, while a proportion $1 - p$ of the population are "jerks" (J), for which $x = 0$ and $y = 1$. Thus we can think that before the game is played, Nature first chooses the type of player 2 that player 1 will meet, according to the probabilities previously described. Draw the game tree.
>
> c. Now assume that for (b) $p = \frac{3}{4}$. What are the pure-strategy Bayesian Nash equilibria of this game? Are there Bayesian Nash equilibria that are not perfect Bayesian equilibria?
>
> d. For which values of p are all the Bayesian Nash equilibria also perfect Bayesian equilibria?

(a) What is asked. (a) The tree and the conditions on x, y under which player 1 trusts; (b) the tree with nice and jerk types; (c) the pure BNE for $p = \frac{3}{4}$ and which are not PBE; (d) the values of p for which every BNE is a PBE.

(b) Method. Backward induction for (a); a Bayesian game (Nature picks player 2's type) for (b)–(d): BNE of the type-contingent normal form, and PBE = BNE in which every type of player 2 also plays optimally at its own, off-path or on-path, node (Tadelis §15.1–15.2, pp. 303–312).

(c) Solution. **(a)** After S player 2 delivers (D, payoff x) or throws away (T, payoff y): he plays D iff $x \geq y$. Player 1 gets 1 after D and -1 after T against 0 from L, so she trusts iff she expects D. Condition for trust in the subgame perfect equilibrium (S, D): $x > y$ (at $x = y$ player 2 is indifferent and she trusts only if he delivers with probability at least $\frac{1}{2}$).

(b) Tree. Nature picks N with probability p ($x = 1$, $y = 0$) or J with $1 - p$ ($x = 0$, $y = 1$); player 1's two nodes form one information set; after S, player 2 moves at the node of his type. Payoffs are (u_1, u_2); after L player 2 gets 0 (he is not involved).



(c) $p = \frac{3}{4}$**.** Type N strictly prefers D ($1 > 0$) and type J strictly prefers T ($1 > 0$), but only if S is played. Let player 2's strategy be (a_N, a_J).

- If S is played, both types must play optimally, so $(a_N, a_J) = (D, T)$ and S pays $\frac{3}{4} - \frac{1}{4} = \frac{1}{2} > 0$: $(S; D, T)$ is a BNE.
- If L is played, the types' actions are off the path, so any (a_N, a_J) is allowed as long as S does not pay more than 0. S yields $2p - 1 = \frac{1}{2}$ against (D, T) and 1 against (D, D): no. It yields -1 against (T, T) and $1 - 2p = -\frac{1}{2}$ against (T, D): yes. So $(L; T, T)$ and $(L; T, D)$ are BNE.

Pure BNE: $(S; D, T)$, $(L; T, T)$, $(L; T, D)$. A PBE needs sequential rationality at the nodes of player 2, which are singletons (each type knows itself): N must play D and J must play T. Only $(S; D, T)$ qualifies, with player 1's belief given by Nature: $\Pr(N) = \frac{3}{4}$. The BNE $(L; T, T)$ and $(L; T, D)$ are not PBE: they rest on the non-credible plan that the nice type throws the problem set away.

(d) For no $p \in [0, 1]$. Whatever p, the profile $(L; T, T)$ is a BNE (after S player 1 would get $-1 < 0$) and it is never a PBE (N would deliver). For $p < \frac{1}{2}$ the PBE is $(L; D, T)$, for $p > \frac{1}{2}$ it is $(S; D, T)$, and for $p = \frac{1}{2}$ both.

(e) Verification. `check_T1.py` enumerates the 2×4 strategy profiles for $p = \frac{3}{4}$ (three BNE, one PBE) and for $p \in \{0, \frac{1}{20}, \ldots, 1\}$ checks that $(L; T, T)$ is a BNE and not a PBE every time. Tadelis prints no answer.

Answer. (a) $x > y$. (c) BNE: $(S; D, T)$, $(L; T, T)$, $(L; T, D)$; the last two are not PBE; the unique PBE is $(S; D, T)$. (d) No value of p: $(L; T, T)$ is a BNE and not a PBE for every p.

Item 4: Muñoz-Garcia Ch. 9, Exercise 1

Foundations · difficulty 2/5

Statement (verbatim from the list, item 4)

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, Ch. 9, Exercise 1 "Finding Separating and Pooling Equilibria", pp. 376–377 [PDF pp. 385–386]. Reproduced verbatim (statement only; the book's worked answer follows it). Answer: yes, the book's worked answer follows the exercise.

> Consider the following game of incomplete information: nature first determines player 1's type, either high or low, with equal probabilities. Then player 1, observing his own type, decides whether to choose left (L) or Right (R). If he chooses left, the game ends and players' payoffs become $(u_1, u_2) = (2, 0)$, regardless of player 1's type. Figure 9.1 depicts these payoffs in the left-hand side of the tree. If, instead, player 1 chooses Right, player 2 is called on to move. In particular, without observing whether player 1's type is high or low, player 2 must respond by either selecting Up (U) or Down (D). As indicated by the payoffs in the right-hand side of the tree, when player 1's type is high, player 2 prefers to play Up. The opposite preference ranking applies when player 1's type is low.

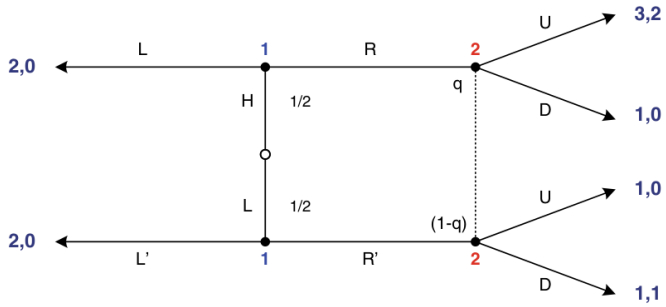


Fig. 9.1 Game tree where player 2 is uninformed

Part (a) Does this game have a Separating Perfect Bayesian Equilibrium (PBE)?

Part (b) Does this game have a Pooling PBE?

(a) What is asked. Whether the game of Fig. 9.1 has (a) a separating PBE and (b) a pooling PBE.

(b) Method. The standard four-step test for each of the four pure profiles (two separating, two pooling): Bayes' rule for the belief, player 2's best reply, then each type's incentive to deviate (Tadelis §16.1, pp. 319–323; Gibbons §4.2.A, pp. 183–190). Payoffs are (u_1, u_2); L gives $(2, 0)$ for both types; after R the high type gets 3 (U) or 1 (D), the low type 1 (U) or 1 (D), and player 2 gets 2 or 0 against H, and 0 or 1 against L for U or D. Let $q = \Pr(H \mid R)$. Player 2's payoffs are $u_2(U) = 2q$ and $u_2(D) = 1 - q$, so U is optimal iff $q \geq \frac{1}{3}$.

(c) Solution. Write a profile as (action of H, action of L) with R/L for the high type and R'/L' for the low type.

Profile	Belief and reply	Deviation check	PBE?
(R, L')	$q = 1$, so U	H: $3 \geq 2$; L-type: $2 \geq 1$	yes
(L, R')	$q = 0$, so D	L-type gets $1 < 2$ from R'	no
(L, L')	q free	H deviates to R if U $(3 > 2)$; no deviation iff D, i.e. $q \leq \frac{1}{3}$	yes, $q \leq \frac{1}{3}$
(R, R')	$q = \frac{1}{2}$, so U	L-type: $1 < 2$ from L'	no

Separating (R, L'). $q = \frac{\frac{1}{2}\cdot 1}{\frac{1}{2}\cdot 1 + \frac{1}{2}\cdot 0} = 1$, so after R player 2 plays U $(2 > 0)$. The high type gets 3 against 2 from deviating to L; the low type gets 2 from L' against 1 from R' answered by U. After L the game ends, so no belief is needed. PBE: $\big((R, L'), U;\ q = 1\big)$.

Separating (L, R'). $q = 0$, player 2 plays D $(1 > 0)$; the low type gets 1 from R' but 2 from deviating to L'. Not an equilibrium.

Pooling (L, L'). R is off the path, so q is free. If $q > \frac{1}{3}$ player 2 would play U and the high type would gain by deviating to R $(3 > 2)$. If $q \leq \frac{1}{3}$ he plays D and both types get $1 < 2$ from deviating: PBE $\big((L, L'), D;\ q \leq \frac{1}{3}\big)$.

Pooling (R, R'). $q = \frac{1}{2} \geq \frac{1}{3}$, so U; the low type gets 1 but 2 from L'. Not an equilibrium.

(e) Verification. `check_T1.py` enumerates all sender and receiver profiles and the beliefs on a grid of step $\frac{1}{60}$: exactly the two PBE above, and the pooling belief set is $q \in [0, \frac{1}{3}]$. The printed answer of the book (pp. 377–385) gives the same two equilibria and the same reasoning for the other two profiles.

Answer. (a) Yes: $\big((R, L'), U;\ q = 1\big)$ is a separating PBE; (L, R') is not. (b) Yes: $\big((L, L'), D;\ q \leq \frac{1}{3}\big)$ is a pooling PBE; (R, R') is not.

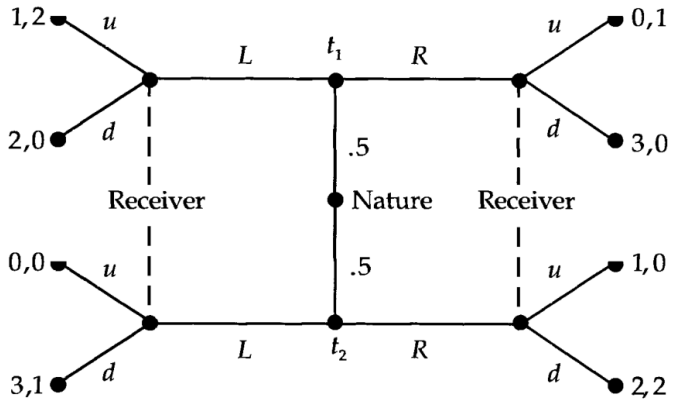
Item 5: Gibbons Problem 4.3

Foundations · difficulty 3/5

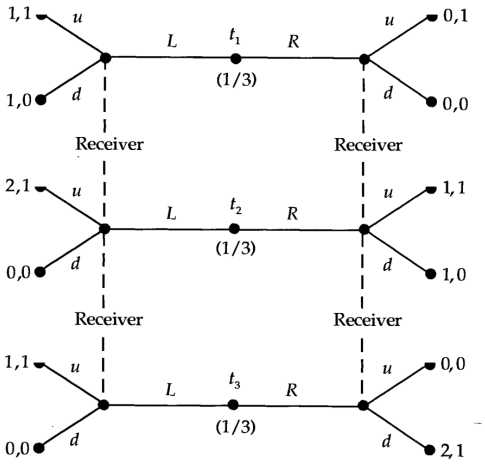
Statement (verbatim from the list, item 5)

Source. Gibbons (1992), *Game Theory for Applied Economists*, §4.2 Problems, Problem 4.3, pp. 246–247 [PDF pp. 252–253]. Reproduced verbatim. Answer: none in the book.

> 4.3. a. Specify a pooling perfect Bayesian equilibrium in which both Sender types play R in the following signaling game.



> b. The following three-type signaling game begins with a move by nature, not shown in the tree, that yields one of the three types with equal probability. Specify a pooling perfect Bayesian equilibrium in which all three Sender types play L.



(a) What is asked. (a) A pooling PBE in which both Sender types play R; (b) a pooling PBE in which all three Sender types play L.

(b) Method. The pooling recipe: (1) belief on the path equals the prior; (2) Receiver's best reply to it; (3) each type's payoff against the best replies after L and after R; (4) off-path belief

and reply that stop every deviation (Gibbons §4.2.A, pp. 183–190, Requirements 1–3; Tadelis §16.1). Payoffs below are (Sender, Receiver), read from the trees.

(c) Solution. **(a)** Types t_1, t_2 with probability $\frac{1}{2}$ each; $\nu = \Pr(t_1 \mid L)$.

	L,u	L,d	R,u	R,d
t_1	$(1,2)$	$(2,0)$	$(0,1)$	$(3,0)$
t_2	$(0,0)$	$(3,1)$	$(1,0)$	$(2,2)$

After R (on the path). Bayes' rule gives $\Pr(t_1 \mid R) = \frac{1}{2}$. The Receiver's payoffs are u: $\frac{1}{2}\cdot 1 + \frac{1}{2}\cdot 0 = \frac{1}{2}$ and d: $\frac{1}{2}\cdot 0 + \frac{1}{2}\cdot 2 = 1$, so he plays d. The types get $u_S(t_1) = 3$ and $u_S(t_2) = 2$.

After L (off the path). The Receiver's payoffs are u: 2ν and d: $1-\nu$, so u is optimal iff $\nu \geq \frac{1}{3}$. If he played d after L (belief $\nu < \frac{1}{3}$), t_2 would get $3 > 2$ from deviating. With u the deviation pays $1 < 3$ to t_1 and $0 < 2$ to t_2.

PBE: both types play R; Receiver plays d after R and u after L; beliefs $\Pr(t_1 \mid R) = \frac{1}{2}$ and $\nu = \Pr(t_1 \mid L) \geq \frac{1}{3}$ (for instance $\nu = \frac{1}{2}$). It is the only pooling-on-R PBE; the off-path reply must be u.

(b) Three types, probability $\frac{1}{3}$ each; $r = (r_1, r_2, r_3)$ is the belief after R.

	L,u	L,d	R,u	R,d
t_1	$(1,1)$	$(1,0)$	$(0,1)$	$(0,0)$
t_2	$(2,1)$	$(0,0)$	$(1,1)$	$(1,0)$
t_3	$(1,1)$	$(0,0)$	$(0,0)$	$(2,1)$

After L (on the path). The belief is $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$; the Receiver's payoff is 1 from u and 0 from d, so he plays u. Payoffs: t_1: 1, t_2: 2, t_3: 1.

After R (off the path). The Receiver's payoffs are u: $r_1 + r_2 + 0 \cdot r_3$ and d: r_3, so u is optimal iff $r_3 \leq \frac{1}{2}$. Against u the deviation pays $0 < 1$, $1 < 2$, $0 < 1$: nobody deviates. Against d the deviation would pay t_3 a payoff of $2 > 1$.

PBE: all types play L; Receiver plays u after L (belief $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$) and u after R, supported by any belief with $r_3 \leq \frac{1}{2}$, for instance $r = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$.

(e) Verification. `check_T1.py` enumerates all pure Sender and Receiver strategies with beliefs on a grid ($\frac{1}{60}$ for (a), $\frac{1}{12}$ for (b)): in (a) the only pooling PBE is RR with replies (u,d) and $\nu \in [\frac{1}{3}, 1]$, with no separating PBE; in (b) the pooling profile is LLL with replies (u,u) and $r_3 \leq \frac{1}{2}$. Gibbons prints no answer.

Answer. (a) Both R; Receiver d after R, u after L; $\Pr(t_1 \mid R) = \frac{1}{2}$, $\Pr(t_1 \mid L) \geq \frac{1}{3}$. (b) All L; Receiver u after L and u after R; belief after R with $r_3 \leq \frac{1}{2}$.

Item 6: Gibbons Problem 4.4

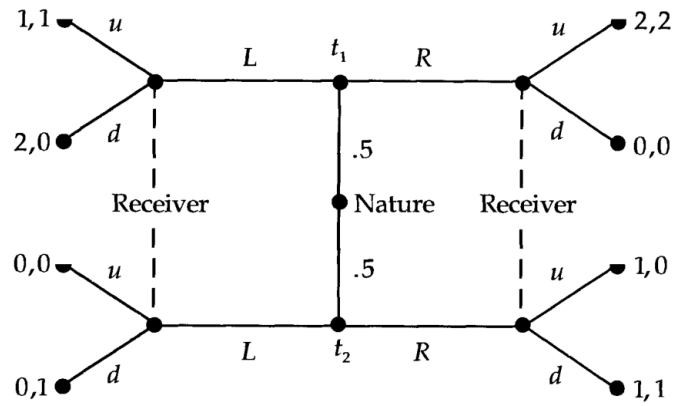
Foundations · difficulty 3/5

Statement (verbatim from the list, item 6)

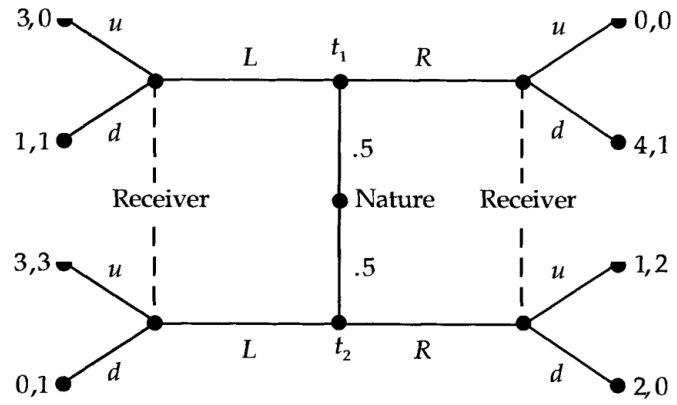
Source. Gibbons (1992), *Game Theory for Applied Economists*, §4.2 Problems, Problem 4.4, p. 248 [PDF p. 254]. Reproduced verbatim. Answer: none in the book.

> 4.4. Describe all the pure-strategy pooling and separating perfect Bayesian equilibria in the following signaling games.

a.



b.



(a) What is asked. All pure-strategy pooling and separating PBE of the two signaling games. Types t_1, t_2 have probability $\frac{1}{2}$ each.

(b) Method. For each of the four Sender profiles (LL, RR, LR, RL, written (message of t_1, message of t_2)): belief from Bayes' rule on the path, free off the path; the Receiver's best reply; each type's payoff against the replies (Gibbons §4.2.A, pp. 183–190; Tadelis §16.1). Payoffs (Sender, Receiver) are read from the trees.

(c) Solution. Game (a).

	L,u	L,d	R,u	R,d
t_1	$(1,1)$	$(2,0)$	$(2,2)$	$(0,0)$
t_2	$(0,0)$	$(0,1)$	$(1,0)$	$(1,1)$

With $\nu = \Pr(t_1 \mid L)$ and $\mu = \Pr(t_1 \mid R)$ the Receiver's payoffs are u: ν against d: $1-\nu$ after L (so u iff $\nu \geq \frac{1}{2}$), and u: 2μ against d: $1-\mu$ after R (so u iff $\mu \geq \frac{1}{3}$).

- LL: t_2 gets 0 after L whatever the reply, but R gives him 1 against either reply. He deviates: not a PBE.
- RR: $\mu = \frac{1}{2} \geq \frac{1}{3}$, reply u; payoffs 2 (t_1) and 1 (t_2). t_2 gets 0 from L. t_1 gets at most 2 from L (with d), which is not more than 2. So RR is a PBE with u after R and any off-path belief ν with the reply best for it (u if $\nu \geq \frac{1}{2}$, d if $\nu \leq \frac{1}{2}$).
- LR: $\nu = 1$, reply u; $\mu = 0$, reply d. t_1 gets 1 (deviating to R: 0); t_2 gets 1 (deviating to L: 0). PBE.
- RL: $\mu = 1$, u; $\nu = 0$, d. t_1 gets 2 (L gives 2, no gain), but t_2 gets 0 from L and 1 from deviating to R (answered by u): not a PBE.

Game (b).

	L,u	L,d	R,u	R,d
t_1	$(3,0)$	$(1,1)$	$(0,0)$	$(4,1)$
t_2	$(3,3)$	$(0,1)$	$(1,2)$	$(2,0)$

The Receiver's payoffs are u: $3(1-\nu)$ against d: 1 after L (so u iff $\nu \leq \frac{2}{3}$), and u: $2(1-\mu)$ against d: μ after R (so u iff $\mu \leq \frac{2}{3}$).

- LL: $\nu = \frac{1}{2}$, reply u; both types get 3. A deviation to R pays t_1 at most 4 (if d) and t_2 at most 2. So the off-path reply must be u ($\mu \leq \frac{2}{3}$): then t_1 gets 0 and t_2 gets 1. PBE with u after L, u after R, $\mu \leq \frac{2}{3}$.
- RR: $\mu = \frac{1}{2}$, reply u; t_1 gets 0, but L pays him 3 (if u) or 1 (if d), both above 0. He deviates: not a PBE.
- LR: $\nu = 1$, reply d; $\mu = 0$, reply u. t_1 gets 1 (deviating to R: 0); t_2 gets 1 (deviating to L: 0). PBE.
- RL: $\mu = 1$, reply d; $\nu = 0$, reply u. t_1 gets 4 (deviating to L: 3); t_2 gets 3 (deviating to R: 2). PBE.

(e) Verification. `check_T1.py` enumerates the 16 Sender–Receiver pure profiles of each game with beliefs on a grid of step $\frac{1}{60}$ and returns exactly the equilibria listed. Gibbons prints no answer.

Answer. (a) Pooling: RR with u after R ($\mu = \frac{1}{2}$) and an off-path reply optimal for ν. Separating: LR with replies (u after L, d after R). No LL, no RL. (b) Pooling: LL with u after L and u after R ($\mu \leq \frac{2}{3}$). Separating: LR (replies d after L, u after R) and RL (u after L, d after R). No RR.

Item 7: Tadelis Exercise 16.5

Foundations · difficulty 3/5

Statement (verbatim from the list, item 7)

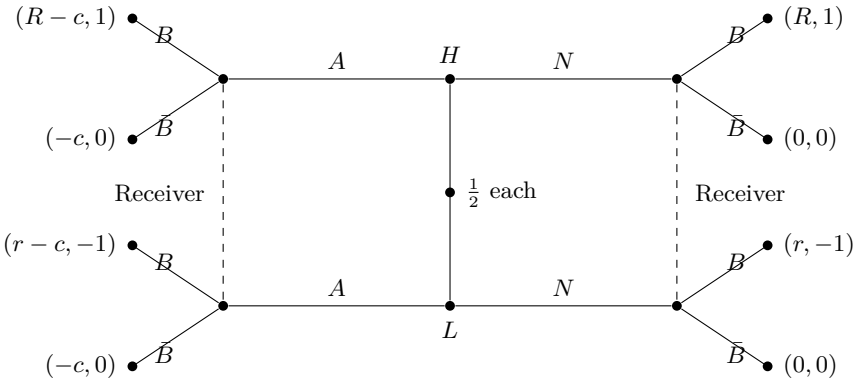
Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 16.5 "Drug Ads", p. 337 [PDF p. 356]. Reproduced verbatim. Answer: none in the book.

> **Drug Ads:** A pharmaceutical company (player 1) introduces a new cold medicine. The medicine may either be highly effective (H) or have little effect (L). The company knows the effectiveness of the drug, but a representative consumer (player 2) knows only that the prior probability that it is highly effective is $\frac{1}{2}$. The company can choose either to advertise the drug excessively (A), at a cost $c > 0$, or not to advertise (N), which costs nothing. The representative consumer decides whether or not to buy the product after observing whether the company advertised the drug. The net payoff to the representative consumer from buying the drug is 1 if it is highly effective and -1 if it has little effect, and his payoff from not buying the drug is 0. If the drug is highly effective then if consumers buy the drug once they will learn of its efficacy and buy it many more times, in which case the company earns a high payoff equal to $R > c$. If instead consumers learn that the drug has little effect then the company will sell the drug to them only once, and the company's returns are equal to $r > 0$. If the representative consumer does not buy the drug then the revenue of the company is 0. Assume that $R > c > r > 0$.
>
> a. Write down the extensive-form game tree.
> b. Find a separating perfect Bayesian Nash equilibrium in which the company chooses a different action depending on the drug's efficacy.
> c. Find a pooling perfect Bayesian Nash equilibrium in which the company chooses the same action regardless of the drug's efficacy.
> d. What changes when $R > r > c > 0$?

(a) What is asked. (a) The extensive form; (b) a separating PBE; (c) a pooling PBE; (d) what changes when $R > r > c > 0$ instead of $R > c > r > 0$.

(b) Method. The five-step method: guess a profile, Bayes' rule, the consumer's reply, the company's incentive check, off-path beliefs (Tadelis §16.1, pp. 319–323). The consumer buys iff his belief that the drug is effective is at least $\frac{1}{2}$: buying pays $\mu - (1-\mu) = 2\mu - 1$ against 0.

(c) Solution. **(a) Tree.** Nature picks H or L with probability $\frac{1}{2}$; the company chooses A (cost c) or N; the consumer, seeing only the message, buys (B) or not ($\bar{B}$). Payoffs (company, consumer): after H: $(R-c, 1)$, $(-c, 0)$ for A and $(R, 1)$, $(0, 0)$ for N; after L: $(r-c, -1)$, $(-c, 0)$ for A and $(r, -1)$, $(0, 0)$ for N.



(The left branches are message A, the right branches message N; upper node = type H, replies B above and $\bar{B}$ below.)

(b) Separating PBE ($R > c > r$). Strategies: H plays A, L plays N. Beliefs: $\mu(A) = 1$, $\mu(N) = 0$. Replies: buy after A ($\mu = 1 \geq \frac{1}{2}$), do not buy after N. Incentives: H gets $R - c > 0$ against 0 from N (since $R > c$); L gets 0 from N against $r - c < 0$ from imitating A (since $c > r$). There are no off-path messages. The reverse profile (H: N, L: A) fails: $\mu(N) = 1$ makes the consumer buy, so L would gain $r > -c$ by switching to N.

(c) Pooling PBE ($R > c > r$). Pooling on A is not an equilibrium: L earns at most $r - c < 0$ (the consumer buys at best, since $\mu(A) = \frac{1}{2}$), while deviating to N earns him at least 0 whatever the consumer does; no off-path belief can prevent this. Pooling on N is: both types play N, $\mu(N) = \frac{1}{2}$, the consumer is indifferent and buys (or randomizes). The types get R and r. A deviation to A gives H at most $R - c < R$ and L at most $r - c < r$, whatever the consumer does after A, so every off-path belief $\mu(A)$ works. PBE: both N; consumer buys after N; after A he best-replies to any belief $\mu(A) \in [0, 1]$. (If he does not buy after N the types earn 0; the profile is still an equilibrium provided he does not buy after A either, i.e. $\mu(A) \leq \frac{1}{2}$.)

(d) $R > r > c > 0$**.** Advertising is now cheaper than the low type's revenue from a sale, so L would mimic: in (A_H, N_L) the consumer buys after A, and L earns $r - c > 0$ from imitating against 0. There is no separating PBE (the reverse profile fails as before). Pooling on N still works, and now pooling on A works too: $\mu(A) = \frac{1}{2}$, the consumer buys, the types get $R - c$ and $r - c > 0$, and a deviation to N is deterred by the off-path belief $\mu(N) < \frac{1}{2}$ (no purchase, payoff $0 < r - c$). Advertising stops separating because it is cheap relative to the low type's gain: single crossing fails.

(e) Verification. `check_T1.py` enumerates all pure profiles for $(R, c, r) = (10, 6, 3)$ and $(10, 2, 3)$ on a grid of beliefs of step $\frac{1}{60}$: the first has the separating PBE (A_H, N_L) and pooling only on N; the second has no separating PBE and pooling on A and on N. Tadelis prints no answer.

Answer. (b) H advertises, L does not; $\mu(A) = 1$, $\mu(N) = 0$; buy after A only. (c) Both N, buy after N, any belief after A (pooling on A is not an equilibrium). (d) No separating PBE; pooling on N and also on A (with $\mu(N) < \frac{1}{2}$ after the off-path N).

Item 8: Muñoz-Garcia Ch. 9, Exercise 2

Foundations · difficulty 2/5

Statement (verbatim from the list, item 8)

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, Ch. 9, Exercise 2 "Job-Market Signaling Game", pp. 385–386 [PDF pp. 394–395] (figure on p. 386). Reproduced verbatim (statement and its footnote 3 only; the book's worked answer follows it). Answer: yes, the book's worked answer follows the exercise.

> Let us consider the sequential game with incomplete information depicted in Fig. 9.11. A worker privately observes whether he has a High productivity or a Low productivity, and then decides whether to acquire some education, such as a college degree, which he will subsequently be able to use as a signal about his innate productivity to potential employers. For simplicity, assume that education is not productivity enhancing. A firm that considers hiring the candidate, however, does not observe the real productivity of the worker, but only knows whether the worker decided to acquire a college education or not (the firm can observe whether the candidate has a valid degree). In this setting, the firm must respond hiring the worker as a manager (M) or as a cashier (C).[3]
>
> **Part (a)** Does this game have an equilibrium using the strategy profile [NE']
>
> **Part (b)** Does this game have an equilibrium using the opposite separating strategy profile [EN']
>
> **Part (c)** Does this game have an equilibrium using the Pooling strategy profile [EE'] in which both types of workers acquire education.
>
> **Part (d)** Does this game have an equilibrium using the pooling strategy profile [NN'], in which neither type of worker acquires education.

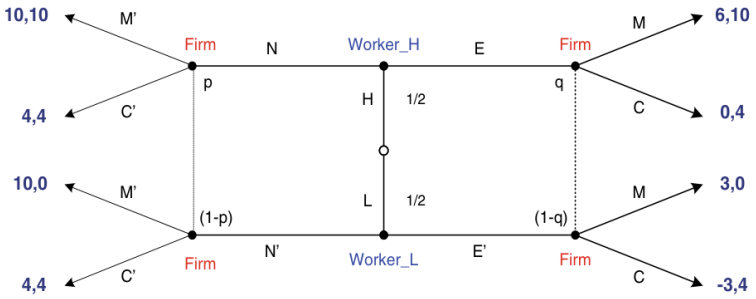


Fig. 9.11 The job-market signaling game

[3]This game presents a simplified version of Spence's (1973) labor market signaling game, where the worker can be of only two types (either high or low productivity), and the firm manager can only respond with two possible actions (either hiring the candidates as a manager, thus offering him a high salary, or as a cashier, at a lower salary). A richer version of this model, with a continuum of education levels and a continuum of salaries, is presented in the next chapter.

(a) What is asked. Whether the job-market signaling game of Fig. 9.11 has an equilibrium in each of the four profiles: (a) $[NE']$, (b) $[EN']$, (c) $[EE']$, (d) $[NN']$ (first letter: the High type's education choice E/N, second: the Low type's E'/N').

(b) Method. Four-step test for each profile: belief by Bayes' rule, the firm's best reply, then each worker type's deviation check. Payoffs (worker, firm): after N the firm gets 10 (M') or 4 (C') from the High type and 0 or 4 from the Low type; the worker gets 10 (M') or 4 (C'). After

E: High worker 6 (M) or 0 (C), Low worker 3 (M) or -3 (C); firm 10 or 4 against High, 0 or 4 against Low. With $p = \Pr(H \mid N)$ and $q = \Pr(H \mid E)$ the firm hires as manager iff $10p \geq 4$ after N and $10q \geq 4$ after E, i.e. $p \geq \frac{2}{5}$, $q \geq \frac{2}{5}$.

(c) Solution.

Profile	Beliefs and replies	Deviation check	PBE?
(a) $[NE']$	$p = 1$, M'; $q = 0$, C	Low: $-3 < 10$ from N'	no
(b) $[EN']$	$q = 1$, M; $p = 0$, C'	High: $6 > 4$; Low: $4 > 3$	yes
(c) $[EE']$	$q = \frac{1}{2}$, M	Low: N' pays 4 or $10 > 3$;	no
(d) $[NN']$	$p = \frac{1}{2}$, M'	no gain from E for any reply	yes

(a) $[NE']$ is not an equilibrium. After N the firm believes High ($p = 1$) and hires a manager; after E it believes Low ($q = 0$) and hires a cashier. The Low worker gets -3 from E' but 10 from deviating to N' (answered by M').

(b) $[EN']$ is an equilibrium. $q = 1$, so after E the firm plays M; $p = 0$, so after N it plays C' ($4 > 0$). High: E gives 6, deviating to N gives 4. Low: N' gives 4, deviating to E' gives 3. PBE: $\big([EN'], (M \text{ after } E,\ C' \text{ after } N);\ q = 1, p = 0\big)$.

(c) $[EE']$ is not an equilibrium. $q = \frac{1}{2} \geq \frac{2}{5}$, so M; the types get 6 and 3. After N the firm answers C' (belief $p < \frac{2}{5}$) or M'. Against C' the Low worker gets $4 > 3$ from N'; against M' both types get 10. Whatever the off-path belief, the Low type deviates.

(d) $[NN']$ is an equilibrium. $p = \frac{1}{2} \geq \frac{2}{5}$, so M' and both types get 10. Any deviation to E gives High at most 6 and Low at most 3, below 10, whatever the firm replies. So the off-path belief $q \in [0, 1]$ is unrestricted. PBE: $\big([NN'], M' \text{ after } N,\ \text{best reply to } q \text{ after } E;\ p = \frac{1}{2}\big)$; the firm answers M if $q \geq \frac{2}{5}$ and C if $q \leq \frac{2}{5}$.

(e) Verification. `check_T1.py` enumerates all pure profiles with beliefs on a grid of step $\frac{1}{60}$: the PBE are exactly $[EN']$ and $[NN']$. The book's printed answer (pp. 386–394) reaches the same verdict on the four profiles.

Answer. (a) No. (b) Yes: $[EN']$ with M after E, C' after N, beliefs $q = 1$, $p = 0$. (c) No. (d) Yes: $[NN']$ with M' after N ($p = \frac{1}{2}$) and any belief q after E with the matching best reply.

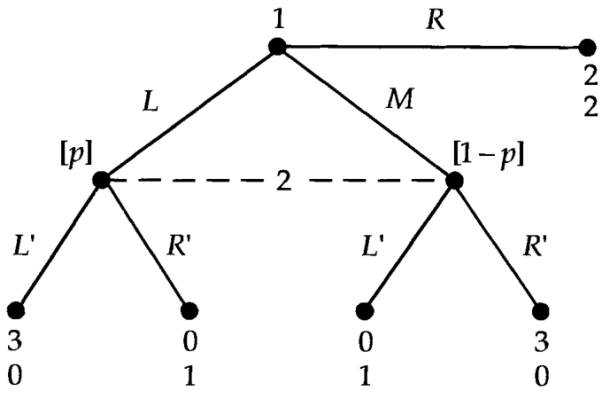
Item 9: Gibbons Problem 4.2

Exam · difficulty 3/5

Statement (verbatim from the list, item 9)

Source. Gibbons (1992), *Game Theory for Applied Economists*, §4.1 Problems, Problem 4.2, pp. 245–246 [PDF pp. 251–252]. Reproduced verbatim. Answer: none in the book.

> 4.2. Show that there does not exist a pure-strategy perfect Bayesian equilibrium in the following extensive-form game. What is the mixed-strategy perfect Bayesian equilibrium?



(a) What is asked. Show that the extensive-form game has no pure-strategy PBE, and find the mixed-strategy PBE.

(b) Method. Player 2's best reply as a function of his belief, then player 1's incentives against each possible reply (Gibbons §4.1, pp. 175–183, Requirements 1–4; a mixed strategy of player 2 requires indifference at his information set, hence a pinned-down belief).

(c) Solution. Payoffs (u_1, u_2): R gives $(2, 2)$; after L: L' gives $(3, 0)$ and R' gives $(0, 1)$; after M: L' gives $(0, 1)$ and R' gives $(3, 0)$. Let p be player 2's belief that L was played, and $q = \Pr(L')$. Player 2's payoffs are

$$u_2(L') = (1-p)\cdot 1 = 1-p, \qquad u_2(R') = p\cdot 1 = p,$$

so L' is optimal iff $p \le \frac{1}{2}$, R' iff $p \ge \frac{1}{2}$, and he mixes only if $p = \frac{1}{2}$. Player 1's payoffs are $u_1(L) = 3q$, $u_1(M) = 3(1-q)$, $u_1(R) = 2$.

No pure PBE. A pure strategy of player 1 and a pure reply of player 2:

- L: then $p = 1$, player 2 plays R', and player 1 gets $0 < 2$ from L. Not an equilibrium.
- M: then $p = 0$, player 2 plays L', and player 1 gets $0 < 2$ from M. Not an equilibrium.
- R: the information set is off the path. If player 2 plays L', player 1 gets $3 > 2$ from deviating to L; if he plays R', $3 > 2$ from deviating to M. Not an equilibrium for any off-path belief.

The mixed PBE. Let player 1 play (a, b, c) on (L, M, R).

- If $a + b > 0$: $p = \frac{a}{a+b}$. For $p < \frac{1}{2}$ player 2 plays L' ($q = 1$) and then only L is a best reply ($3 > 0$ and $3 > 2$), so $b = 0$ and $p = 1$: contradiction. For $p > \frac{1}{2}$ symmetric. For $p = \frac{1}{2}$, $a = b$ and player 2 mixes, so indifference of player 1 between L and M forces $q = \frac{1}{2}$ and payoff $\frac{3}{2} < 2 = u_1(R)$, so $a = b = 0$: contradiction.
- So $c = 1$ (player 1 plays R for sure). The belief is not pinned down by Bayes' rule; for player 2 to mix it must be $p = \frac{1}{2}$, and he mixes with q such that player 1 does not deviate: $3q \leq 2$ and $3(1 - q) \leq 2$.

The mixed PBE is: player 1 plays R; player 2 plays L' with probability $q \in [\frac{1}{3}, \frac{2}{3}]$ and R' with probability $1 - q$; belief $p = \frac{1}{2}$ at his information set. Player 1 earns 2, and L and M earn at most $3q \leq 2$ and $3(1 - q) \leq 2$. This is the only family of PBE.

(e) Verification. `check_T2.py` enumerates player 1's mixtures and player 2's q on a grid of step $\frac{1}{12}$, with beliefs fixed by Bayes' rule where it applies and free elsewhere: the only equilibria have $c = 1$, $p = \frac{1}{2}$ and $q \in \{\frac{4}{12}, \ldots, \frac{8}{12}\}$. Gibbons prints no answer.

Answer. No pure-strategy PBE. The (unique) mixed PBE family: player 1 plays R; player 2 plays L' with probability $q \in [\frac{1}{3}, \frac{2}{3}]$ and R' otherwise; belief $\Pr(L) = \frac{1}{2}$.

Item 10: Tadelis Exercise 15.5

Exam · difficulty 4/5

Statement (verbatim from the list, item 10)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 15.5 "Entry (again)", pp. 316–317 [PDF pp. 335–336]. Reproduced verbatim. Answer: none in the book.

> **Entry (again):** The market for widgets has an incumbent firm. The total value of having 100% of the market is 10, which the incumbent receives if no one enters. A potential entrant arrives, and it can be one of two types: tough (T) or weak (W). A weak entrant can choose one of three options: small entry (S), big entry (B), or exit (X). A tough entrant can choose only between S or B (it is inconceivable that it would choose X). There is no cost for a tough entrant to enter at any level. However, it costs a weak entrant 6 to enter at any level. Exiting costs nothing to the entrant. The entrant knows his type, but the incumbent knows only the prior distribution: $\Pr\{T\} = \frac{1}{2}$.
>
> In response to any level of entry, the incumbent can choose to accommodate (A) or fight (F). Accommodating an entrant imposes no costs. Independent of the entrant's type, accommodating small entry gives the incumbent 60% of the market and the entrant 40%, while accommodating big entry gives the incumbent 40% of the market and the entrant 60%. Fighting a tough entrant increases the incumbent's market share by 20% (relative to accommodating) but imposes a cost of 4 on the incumbent. Fighting a weak entrant that chose S increases the market share of the incumbent to 100% but imposes a cost of 2 on the incumbent. Fighting a weak entrant that chose B increases the market share of the incumbent to 100% but imposes a cost of 8 on the incumbent.
>
> a. Draw this game in extensive form.
> b. Using a matrix representation, find all the pure-strategy Bayesian Nash equilibria for this game.
> c. Which one of the Bayesian Nash equilibria is preferred by the incumbent? Can it be supported as a perfect Bayesian equilibrium?
> d. Find all the perfect Bayesian equilibria of this game.

(a) What is asked. (a) The extensive form; (b) all pure-strategy BNE by matrix; (c) the BNE the incumbent prefers and whether it is a PBE; (d) all PBE.

(b) Method. Compute the terminal payoffs, build the Bayesian-game matrix, read off the BNE, then add beliefs and sequential rationality for the incumbent at each of his two information sets (Tadelis §15.1–15.2, pp. 303–312; §16.1).

(c) Solution. **Payoffs.** The market is worth 10 (shares are multiplied by 10). A weak entrant pays 6 to enter at any level. Entries are (entrant, incumbent):

Type	Entry	Incumbent accommodates (A)	Incumbent fights (F)
T	S	$(4,\ 6)$	$(2,\ 8-4=4)$
T	B	$(6,\ 4)$	$(4,\ 6-4=2)$
W	S	$(4-6=-2,\ 6)$	$(-6,\ 10-2=8)$
W	B	$(6-6=0,\ 4)$	$(-6,\ 10-8=2)$
W	X	$(0,\ 10)$	

(a) Tree. Nature picks T or W with probability $\frac{1}{2}$; the entrant chooses S or B (T) or S, B or X (W); X ends the game; the incumbent has two information sets, one after S (two nodes) and one after B (two nodes), and chooses A or F. Terminal payoffs are in the table.

(b) BNE. Entrant strategies are (entry of T, entry of W), e.g. SX; incumbent strategies are (reply after S, reply after B). Expected payoffs (entrant, incumbent), best replies underlined:

	AA	AF	FA	FF
SS	1, $\underline{6}$	1, $\underline{6}$	-2, $\underline{6}$	-2, $\underline{6}$
SB	2, $\underline{5}$	-1, 4	1, 4	-2, 3
SX	2, $\underline{8}$	$\underline{2}$, $\underline{8}$	1, 7	1, 7
BS	2, 5	1, 4	0, $\underline{6}$	-1, 5
BB	$\underline{3}$, $\underline{4}$	-1, 2	$\underline{3}$, $\underline{4}$	-1, 2
BX	$\underline{3}$, $\underline{7}$	$\underline{2}$, 6	$\underline{3}$, $\underline{7}$	$\underline{2}$, 6

Pure BNE: (SX, AF), (BB, AA), (BB, FA), (BX, AA), (BX, FA).

(c) The incumbent's payoffs in the five BNE are 8, 4, 4, 7, 7. The preferred one is (SX, AF): the tough type enters small, the weak type stays out, and the incumbent accommodates S and fights B. It is not a PBE: after B the incumbent gets 4 from A and 2 from F against *both* types, so A is better for every belief; the threat to fight B is not credible.

(d) PBE. Let μ_W be the belief that the entrant is weak after S. After B the incumbent plays A ($4 > 2$ against either type). After S he gets 6 from A and $4(1-\mu_W)+8\mu_W$ from F, so he fights iff $\mu_W \geq \frac{1}{2}$. Given A after B: the tough type earns 6 from B and at most 4 from S, so he plays B; the weak type earns 0 from B and X, and at most -2 from S, so he plays B or X. Both profiles are equilibria, and the off-path reply after S can be A (with $\mu_W \leq \frac{1}{2}$) or F (with $\mu_W \geq \frac{1}{2}$), since neither type gains by deviating to S:

Entrant	Incumbent	Beliefs
T: B, W: B	A after B; A or F after S	$\Pr(W \mid B) = \frac{1}{2}$; after S matching μ_W
T: B, W: X	A after B; A or F after S	$\Pr(W \mid B) = 0$; after S matching μ_W

Incumbent's payoffs: 4 in the first (both enter big), 7 in the second (the weak type exits). Pooling on S is not an equilibrium (the weak type would exit), and neither is the profile in which the tough type enters small (it would switch to B).

(e) Verification. `check_T2.py` builds the 6×4 matrix from the payoff table, lists the five BNE, enumerates PBE with beliefs on a grid of step $\frac{1}{60}$ (the four PBE above, two outcomes), and checks the threshold $\mu_W \geq \frac{1}{2}$ after S. Tadelis prints no answer.

Answer. (b) BNE: (SX, AF), (BB, AA), (BB, FA), (BX, AA), (BX, FA). (c) The incumbent prefers (SX, AF) (payoff 8); it is not a PBE because F after B is never optimal. (d) PBE: the tough type enters big; the weak type enters big or exits; the incumbent accommodates after B and responds after S according to $\mu_W \lessgtr \frac{1}{2}$.

Item 11: Muñoz-Garcia Ch. 9, Exercise 5

Exam · difficulty 3/5

Statement (verbatim from the list, item 11)

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, Ch. 9, Exercise 5 "Finding Semi-Separating Equilibrium", pp. 422–423 [PDF pp. 431–432]. Reproduced verbatim (statement only; the book's worked answer follows it). "Obviously, player makes his choice" is as printed. Answer: yes, the book's worked answer follows the exercise.

> Consider the sequential game with incomplete information depicted in Fig. 9.46. Two players are playing the following super-simple poker game. The first player gets his cards, and obviously he is the only one who can observe them. He can either get a High hand or a Low hand, with equal probabilities. After observing his hand, he must decide whether to Bet (let us imagine that he is betting a fixed amount of a dollar) or Resign from the game. If he resigns his payoff is -1, independent of whether he has a High or a Low hand, and player 2 gets a dollar. If he bets, player 2 must decide whether to Call or Fold from the game. Obviously, player makes his choice without being able to observe player 1's hands, but only observing that player 1 Bets.

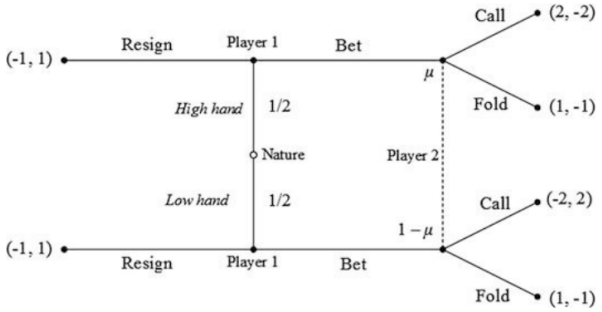


Fig. 9.46 A simple poker game

(a) Show that no separating PBE can be sustained where player 1 chooses a pure strategy (i.e., betting or resigning, given his cards, with 100% probability).

(b) Show that no pooling PBE can be sustained where player 1 chooses a pure strategy (i.e., betting or resigning, given his cards, with 100% probability).

(c) Find a semi-separating PBE where player 1 chooses Bet or Resign in pure strategies when having a high hand, but randomizes between bet and resign when having a low hand.

(a) What is asked. In the poker game of Fig. 9.46: (a) no pure separating PBE; (b) no pure pooling PBE; (c) a semi-separating PBE in which the High hand bets and the Low hand mixes.

(b) Method. Pure profiles are refuted by one deviation each; the semi-separating PBE comes from two indifference conditions (player 2's belief makes him indifferent between Call and Fold, and his mixing makes the Low hand indifferent between Bet and Resign) plus Bayes' rule (Tadelis §16.1; Gibbons §4.2.A). Payoffs (u_1, u_2): Resign $(-1, 1)$; Bet and Call $(2, -2)$ with the High hand and $(-2, 2)$ with the Low hand; Bet and Fold $(1, -1)$ for both. Let $\mu = \Pr(\text{High} \mid \text{Bet})$. Player 2's payoffs are $u_2(\text{Call}) = -2\mu + 2(1-\mu) = 2 - 4\mu$ and $u_2(\text{Fold}) = -1$, so he calls iff $\mu \leq \frac{3}{4}$.

(c) Solution. **(a) Separating.** *High bets, Low resigns:* $\mu = 1$, so Fold. The Low hand gets -1 from resigning but 1 from betting against a Fold: he deviates. *High resigns, Low bets:* $\mu = 0$, so Call. The Low hand gets -2 from betting but -1 from resigning: he deviates (so does the High hand: -1 from resigning against 2 from betting and being called).

(b) Pooling. *Both bet:* $\mu = \frac{1}{2}$, $2 - 4\mu = 0 > -1$, so Call. The Low hand gets $-2 < -1$ and deviates to Resign. *Both resign:* Bet is off the path. The High hand gets 2 (Call) or 1 (Fold), both above -1, so he deviates whatever the off-path belief.

(c) Semi-separating. Let the High hand bet, the Low hand bet with probability β, and player 2 call with probability γ.

(i) *Beliefs (Bayes).* $\mu = \dfrac{\frac{1}{2} \cdot 1}{\frac{1}{2} \cdot 1 + \frac{1}{2}\beta} = \dfrac{1}{1+\beta}$.

(ii) *Player 2 mixes* only if indifferent: $2 - 4\mu = -1$, i.e. $\mu = \frac{3}{4}$. Then $1 + \beta = \frac{4}{3}$, so $\beta = \frac{1}{3}$.

(iii) *The Low hand mixes* only if indifferent between Bet and Resign: $-2\gamma + 1 \cdot (1 - \gamma) = -1$, i.e. $1 - 3\gamma = -1$ and $\gamma = \frac{2}{3}$.

(iv) *The High hand* gets $2\gamma + (1 - \gamma) = \frac{5}{3} > -1$ from betting, so betting is strictly optimal.

PBE: the High hand bets; the Low hand bets with probability $\frac{1}{3}$ and resigns with $\frac{2}{3}$; player 2 calls with probability $\frac{2}{3}$; belief $\mu = \frac{3}{4}$. Payoff of player 1 with the Low hand: -1; with the High hand: $\frac{5}{3}$. The mixing probability β is unique: any other β gives $\mu \neq \frac{3}{4}$, so player 2 would play a pure reply and the Low hand would not be indifferent.

(e) Verification. `check_T2.py` enumerates all pure sender and receiver profiles (none is a PBE), and checks exactly that $\mu = \frac{3}{4}$, that Call and Fold both give -1, that the Low hand's Bet and Resign both give -1 and that the High hand's bet gives $\frac{5}{3}$. The book's printed answer reaches $\mu = \frac{3}{4}$ and a Call probability of 66.67%, in agreement.

Answer. (a), (b): no pure separating or pooling PBE. (c) The High hand bets; the Low hand bets with probability $\frac{1}{3}$; player 2 calls with probability $\frac{2}{3}$; $\mu = \frac{3}{4}$.

Item 12: Tadelis Exercise 16.4

Exam · difficulty 3/5

Statement (verbatim from the list, item 12)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 16.4 "School Choice", pp. 336–337 [PDF pp. 355–356]. Reproduced verbatim. Answer: none in the book.

> **School Choice:** A prospective student, player 1, is deciding whether to go to Brown Bear University (B) or Silly Tree University (S). Both are top-rated universities that accept only the very best students. The difference is that Silly Tree offers a more holistic and comfortable lifestyle, while Brown Bear requires harder work and imposes more "studying" costs on its students. For this reason a player who goes to Brown Bear learns to become more self-sufficient and becomes more productive in the workforce. The cost of learning and level of final productivity depend on the type of player 1, who can be either excellent (E) or just really good (G) (because no other type would be accepted into these universities). Player 1 knows his type, but all others in society know only that a proportion p of young adults are type E. The cost of learning and the level of productivity from each choice are given as follows:

Type	University choice	Learning cost	Productivity
E	B	2	12
E	S	0	4
G	B	8	10
G	S	2	2

> Once player 1 finishes school, he is hired by a firm (player 2), which can place him in one of two jobs: low-tech (L) or high-tech (H). The wage for the L job is $w_L = 2$ and that for the H job is $w_H = 6$. The payoffs to player 1 are the wages less the cost of education. The firm's profits depend on both the job assignment and the type of employee. If the employee is assigned to an H job, the net profits to the firm are equal to the productivity of the employed player 1 less the wage he is paid. If the assignment is to an L job, the net profits are half the productivity of the employed player 1 less the wage he is paid.
>
> a. Draw this game in extensive form.
> b. Assume that $p = \frac{1}{2}$. Represent the matrix form of the Bayesian game.
> c. Find all the pure-strategy Bayesian Nash equilibria.
> d. Find all the pure-strategy perfect Bayesian equilibria.
> e. In brief terms, what is the intuition that explains the comparison between your results in (d) and (c)?

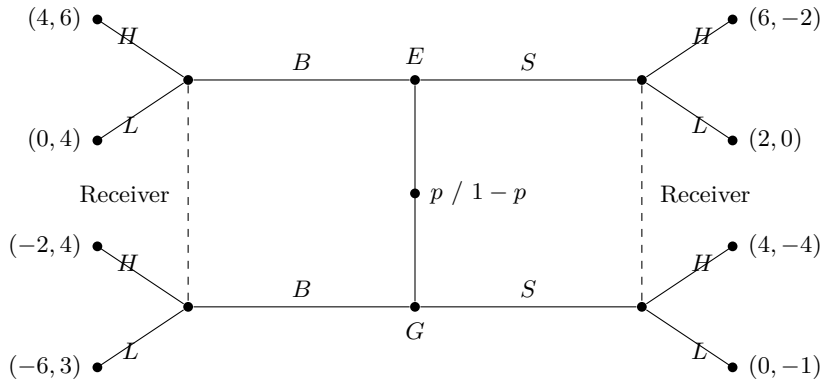
(a) What is asked. (a) The extensive form; (b) the matrix of the Bayesian game for $p = \frac{1}{2}$; (c) all pure-strategy BNE; (d) all pure-strategy PBE; (e) the intuition for the difference.

(b) Method. Compute payoffs from the table (wage minus learning cost for the student; productivity minus wage for the firm, with the L job worth half the productivity); matrix and BNE; PBE by the firm's best replies at its two information sets (Tadelis §16.1, pp. 319–323).

(c) Solution. **Payoffs** (student, firm); wages $w_L = 2$, $w_H = 6$:

Type	School	Firm assigns H	Firm assigns L
E	B	$(6-2,\ 12-6) = (4, 6)$	$(2-2,\ 6-2) = (0, 4)$
E	S	$(6-0,\ 4-6) = (6, -2)$	$(2-0,\ 2-2) = (2, 0)$
G	B	$(6-8,\ 10-6) = (-2, 4)$	$(2-8,\ 5-2) = (-6, 3)$
G	S	$(6-2,\ 2-6) = (4, -4)$	$(2-2,\ 1-2) = (0, -1)$

(a) Tree. Nature picks E (probability p) or G; the student chooses B or S; the firm observes only the school and chooses H or L.



(Left branches: B; right branches: S; upper node: type E.)

(b) Matrix, $p = \frac{1}{2}$. The student's strategy is (school of E, school of G); the firm's is (job after B, job after S). Entries (student, firm):

	HH	HL	LH	LL
BB	$1,\ \underline{5}$	$1,\ \underline{5}$	$-3,\ \frac{7}{2}$	$-3,\ \frac{7}{2}$
BS	$4,\ 1$	$\underline{2},\ \underline{\frac{5}{2}}$	$2,\ 0$	$0,\ \frac{3}{2}$
SB	$2,\ 1$	$0,\ \underline{2}$	$0,\ \frac{1}{2}$	$-2,\ \frac{3}{2}$
SS	$\underline{5},\ -3$	$1,\ \underline{-\frac{1}{2}}$	$\underline{5},\ -3$	$\underline{1},\ \underline{-\frac{1}{2}}$

For instance (BS, HL): E goes to B and is placed in H (4), G goes to S and is placed in L (0): student $\frac{1}{2}4 + \frac{1}{2}0 = 2$, firm $\frac{1}{2}6 + \frac{1}{2}(-1) = \frac{5}{2}$.

(c) BNE. The cells with both payoffs underlined: (BS, HL) and (SS, LL).

(d) PBE. The firm's replies are dominant at each information set: after B it gets $6 > 4$ (against E) and $4 > 3$ (against G) from H, so H is optimal for every belief; after S it gets $0 > -2$ (against E) and $-1 > -4$ (against G) from L, so L is optimal for every belief and every p. Hence in any PBE the firm plays (H after B, L after S). Then E prefers B ($4 > 2$) and G prefers S ($0 > -2$): the unique PBE is (BS, HL), separating, with beliefs $\Pr(E \mid B) = 1$, $\Pr(E \mid S) = 0$ (Bayes' rule at both sets). The firm's payoff is $\frac{5}{2}$.

(e) Intuition. (SS, LL) is a BNE only because the firm's reply L to the unreached message B is never tested: it is a threat that no belief makes optimal (H strictly dominates L after B), and it keeps E from signaling. A BNE accepts non-credible off-path replies; a PBE requires the reply to be optimal at every information set for some belief.

(e) Verification. `check_T2.py` recomputes the eight payoff pairs from wages, costs and productivities, builds the matrix, lists the BNE (BS, HL), (SS, LL), enumerates PBE (only (BS, HL)), and checks the dominance of H after B and L after S for $p \in \{0, \frac{1}{20}, \ldots, 1\}$. Tadelis prints no answer.

Answer. (c) BNE: (BS, HL) and (SS, LL). (d) The only PBE is (BS, HL): E goes to Brown Bear, G to Silly Tree, the firm assigns H after B and L after S (for every p). (e) (SS, LL) relies on a non-credible reply to B.

Item 13: Tadelis Exercise 16.3

Exam · difficulty 3/5

Statement (verbatim from the list, item 13)

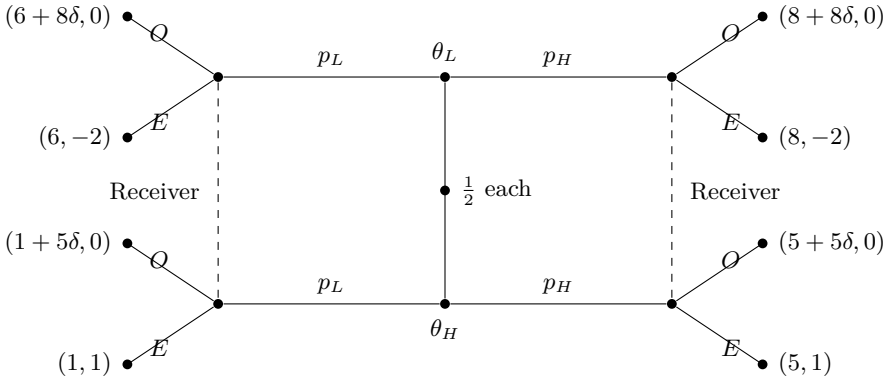
Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 16.3 "More Limit Pricing", pp. 335–336 [PDF pp. 354–355]. Reproduced verbatim. Answer: none in the book.

> **More Limit Pricing:** An incumbent firm (player 1) is either a low-cost type $\theta_1 = \theta_L$ or a high-cost type $\theta_1 = \theta_H$, each with equal probability. In period $t = 1$ the incumbent is a monopolist and sets one of two prices p_L or p_H, and its profits in this period depend on its type and the price it chooses, given by the following table:
>
> | Type | Profit from p_L | Profit from p_H |
> |---|---|---|
> | θ_L | 6 | 8 |
> | θ_H | 1 | 5 |
>
> After observing the period $t = 1$ price, a potential entrant (player 2), which does not know the incumbent's type but knows the distribution of types, can choose to enter the market (E) or stay out (O) in period $t = 2$. The payoffs of both players in period 2 depend on the entrant's choice and on the incumbent's type and are given by the following table:
>
> | Incumbent's type | Entrant's choice | Incumbent's payoff | Entrant's payoff |
> |---|---|---|---|
> | θ_L | E | 0 | -2 |
> | θ_L | O | 8 | 0 |
> | θ_H | E | 0 | 1 |
> | θ_H | O | 5 | 0 |
>
> At the beginning of the game the incumbent discounts profits for period $t = 2$ using a discount factor $\delta \leq 1$.
>
> a. Draw the extensive-form game tree of this game and write down the corresponding matrix.
> b. For $\delta = 1$ find a pooling perfect Bayesian equilibrium of the game in which both types of player 1 choose p_L in period $t = 1$.
> c. Find the range of discount factors for which a separating perfect Bayesian equilibrium of the game exists in which type θ_L chooses p_L and type θ_H chooses p_H in period $t = 1$.

(a) What is asked. (a) The tree and the matrix; (b) for $\delta = 1$, a pooling PBE in which both types choose p_L; (c) the discount factors for which a separating PBE with θ_L at p_L and θ_H at p_H exists.

(b) Method. Compute the entrant's entry rule from his belief, then the two incentive constraints of a separating or pooling profile (Tadelis §16.2, pp. 323–332: same steps, with a discrete price). With belief $\mu = \Pr(\theta_H \mid \text{price})$ the entrant gets $1 \cdot \mu - 2(1 - \mu) = 3\mu - 2$ from E and 0 from O: he enters iff $\mu \geq \frac{2}{3}$.

(c) Solution. **(a) Tree and matrix.** Nature picks θ_L or θ_H with probability $\frac{1}{2}$; the incumbent picks p_L or p_H; the entrant sees the price and picks E or O. Incumbent's payoff: period-1 profit plus δ times period-2 payoff; entrant's payoff: period 2 only.



(Left branches p_L, right branches p_H; upper node θ_L; replies O above, E below; payoffs (incumbent, entrant).) The matrix, with incumbent strategies (price of θ_L, price of θ_H) and entrant strategies (reply to p_L, reply to p_H):

	OO	OE	EO	EE
p_Lp_L	$\frac{7}{2}+\frac{13}{2}\delta,\ 0$	$\frac{7}{2}+\frac{13}{2}\delta,\ 0$	$\frac{7}{2},\ -\frac{1}{2}$	$\frac{7}{2},\ -\frac{1}{2}$
p_Lp_H	$\frac{11}{2}+\frac{13}{2}\delta,\ 0$	$\frac{11}{2}+4\delta,\ \frac{1}{2}$	$\frac{11}{2}+\frac{5}{2}\delta,\ -1$	$\frac{11}{2},\ -\frac{1}{2}$
p_Hp_L	$\frac{9}{2}+\frac{13}{2}\delta,\ 0$	$\frac{9}{2}+\frac{5}{2}\delta,\ -1$	$\frac{9}{2}+4\delta,\ \frac{1}{2}$	$\frac{9}{2},\ -\frac{1}{2}$
p_Hp_H	$\frac{13}{2}+\frac{13}{2}\delta,\ 0$	$\frac{13}{2},\ -\frac{1}{2}$	$\frac{13}{2}+\frac{13}{2}\delta,\ 0$	$\frac{13}{2},\ -\frac{1}{2}$

(b) Pooling at p_L, $\delta = 1$. After p_L the belief is the prior, $\mu = \frac{1}{2} < \frac{2}{3}$, so the entrant stays out (O). Payoffs: θ_L: $6 + 8 = 14$; θ_H: $1 + 5 = 6$. Off the path (after p_H) choose the belief $\mu \geq \frac{2}{3}$ (say $\mu = 1$): the entrant enters, and a deviation to p_H pays θ_L $8 + 0 = 8 < 14$ and θ_H $5 + 0 = 5 < 6$. PBE: both types p_L; the entrant plays O after p_L ($\mu = \frac{1}{2}$) and E after p_H ($\mu \geq \frac{2}{3}$). (The belief after p_H cannot be below $\frac{2}{3}$: then he would stay out and θ_L would get $16 > 14$.)

(c) Separating. Beliefs: $\mu(p_L) = 0$, $\mu(p_H) = 1$; replies: O after p_L, E after p_H (Bayes' rule on both prices, so there are no off-path prices). Incentive constraints:

$$\begin{aligned} \theta_L: &\quad 6 + 8\delta \geq 8 + 0 \iff \delta \geq \tfrac{1}{4}, \\ \theta_H: &\quad 5 + 0 \geq 1 + 5\delta \iff \delta \leq \tfrac{4}{5}. \end{aligned}$$

θ_L must not give up the high period-1 profit of p_H (8 against 6) only to be mistaken for θ_H; θ_H must not mimic ($1 + 5\delta$ against 5). Both hold iff $\frac{1}{4} \leq \delta \leq \frac{4}{5}$. (The reverse separating profile never works: θ_H would get 1 from p_L and $5 + 5\delta$ from p_H.)

(e) Verification. `check_T2.py` builds the game for each $\delta \in \{0, \frac{1}{40}, \ldots, 1\}$ and enumerates pure PBE with beliefs on a grid of step $\frac{1}{60}$: the separating profile exists exactly for $\frac{1}{4} \leq \delta \leq \frac{4}{5}$; at $\delta = 1$ the profile of (b) is a PBE with off-path belief at least $\frac{2}{3}$. Pooling at p_H also exists for every δ, which the exercise does not ask. Tadelis prints no answer.

Answer. (b) Both p_L; O after p_L (belief $\frac{1}{2}$), E after p_H (belief $\mu \geq \frac{2}{3}$). (c) $\frac{1}{4} \leq \delta \leq \frac{4}{5}$.

Item 14: Muñoz-Garcia Ch. 11, Exercise 5

Exam · difficulty 4/5

Statement (verbatim from the list, item 14)

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, Ch. 11, Exercise 5 "Entry Deterrence Through Price Wars", p. 505 [PDF p. 512]. Reproduced verbatim (statement only; the book's worked answer follows it). Answer: yes, the book's worked answer follows the exercise.

> Consider two firms competing in an industry, an incumbent and an entrant. The entrant has already entered the market, and in period 1 the incumbent must decide whether to fight or accommodate; if she fights, the entrant loses $f_E > 0$ (i.e., profit $= -f_E$) and if she accommodates the entrant makes a profit of $a_E > 0$. The incumbent may be either "sane", in which case he loses $f_I > 0$ from fighting, and gains $a_I > 0$ from accommodating; or "crazy", in which case his only available action is to fight. In period 2, the entrant must choose whether to stay in the market or exit; if she exits, the incumbent makes a profit of m; if she stays, the incumbent gets to fight or accommodate again and the payoffs are as described before. Assume that there is no discounting, and that the prior probability of a "sane" incumbent is $0 < p < 1$ (the actual type of the incumbent is private knowledge to the incumbent).
>
> **Part (a)** Draw the game in its extensive form.
>
> **Part (b)** Show that if payoffs satisfy $m - f_I \leq 2a_I$, then there is a separating equilibrium.
>
> **Part (c)** Show that if $m - f_I \leq 2a_I$ does not hold there is a pooling equilibrium when $p \leq \frac{f_E}{f_E + a_E}$.

(a) What is asked. (a) The extensive form; (b) if $m - f_I \leq 2a_I$ there is a separating equilibrium; (c) if $m - f_I \leq 2a_I$ fails, there is a pooling equilibrium when $p \leq \frac{f_E}{f_E + a_E}$.

(b) Method. Backward induction in the last stage, then a signaling game between the incumbent (sender, sane or crazy) and the entrant (receiver) with the first-period action as message (Tadelis §16.1, pp. 319–323; the structure is Kreps–Wilson/Milgrom–Roberts reputation for toughness, §16.2).

(c) Solution. **(a) Tree.** Nature picks *sane* (probability p) or *crazy* ($1 - p$). In period 1 the sane incumbent chooses Fight (F_1) or Accommodate (Acc_1); the crazy one can only fight. The entrant observes this (not the type) and chooses Stay or Exit. After Exit the incumbent earns m in period 2; after Stay the incumbent chooses again (the crazy type fights) and the period-2 payoffs are as in period 1. Totals (incumbent, entrant), with the sane incumbent's period-2 choice already optimized:

Incumbent	Period 1	Stay (then sane plays Acc_2)	Exit
sane	Acc_1	$(2a_I,\ 2a_E)$	$(a_I + m,\ a_E)$
sane	F_1	$(a_I - f_I,\ a_E - f_E)$	$(m - f_I,\ -f_E)$
crazy	F_1	$(\cdot,\ -2f_E)$	$(\cdot,\ -f_E)$

The sane incumbent plays Acc_2 in the last stage after either history: $a_I > -f_I$ gives $a_I - f_I > -2f_I$ after F_1, and $2a_I > a_I - f_I$ after Acc_1. The entrant's information sets: after Acc_1 (only the sane type reaches it) and after F_1 (both types; belief $\lambda = \Pr(\text{sane} \mid F_1)$).

Entrant's replies. After Acc_1 the entrant knows he faces the sane type: Stay gives $2a_E > a_E$ (Exit). After F_1: Stay gives $\lambda(a_E - f_E) + (1-\lambda)(-2f_E)$, Exit gives $-f_E$, so he stays iff

$$\lambda(a_E - f_E) - 2(1-\lambda)f_E \geq -f_E \iff \lambda \geq \frac{f_E}{f_E + a_E} =: \lambda^*.$$

(b) Separating. The sane type accommodates, the crazy type fights. Bayes' rule: after F_1, $\lambda = 0 < \lambda^*$, so the entrant exits; after Acc_1 he stays. The sane type gets $2a_I$ from Acc_1 and, by deviating to F_1 (answered by Exit), $m - f_I$. Hence the profile is a PBE iff $m - f_I \leq 2a_I$. As the crazy type has no choice, this is the only separating equilibrium.

(c) Pooling on fight. Suppose $m - f_I > 2a_I$ and both types fight, so $\lambda = p$.

- If $p \leq \lambda^*$ the entrant exits. The sane type gets $m - f_I$ from F_1. A deviation to Acc_1 is met by Stay (belief 1 on the sane type: it is a one-node information set) and pays $2a_I < m - f_I$. So pooling is a PBE.
- If $p > \lambda^*$ the entrant stays and the sane type gets $a_I - f_I < 2a_I$ from F_1: he deviates to Acc_1. No pooling equilibrium.

So when $m - f_I \leq 2a_I$ fails there is a pooling equilibrium iff $p \leq \frac{f_E}{f_E + a_E}$: the entrant must be pessimistic enough about a sane incumbent to exit after a fight, which then makes fighting worth its cost to the sane type. (At $p = \lambda^*$ he is indifferent and exits.)

(e) Verification. `check_T2.py` solves the game for 400 random integer parameter sets and rational priors, enumerating all pure PBE with the belief after Acc_1 restricted to the sane type: a separating PBE exists exactly when $m - f_I \leq 2a_I$, and a pooling-on-fight PBE exactly when $m - f_I \geq 2a_I$ and $p \leq \frac{f_E}{f_E + a_E}$. The book's printed answer (pp. 505–510) states the same conditions, including the observation that the separating equilibrium is the only one.

Answer. (b) Sane incumbent accommodates, crazy fights; entrant exits after a fight and stays after an accommodation; a PBE iff $m - f_I \leq 2a_I$. (c) Both fight, $\lambda = p$; entrant exits after a fight (needs $p \leq \frac{f_E}{f_E + a_E}$) and stays after an accommodation; a PBE when $m - f_I \geq 2a_I$.

Item 15: Tadelis Exercise 16.6

Exam · difficulty 3/5

Statement (verbatim from the list, item 15)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 16.6 "Publish or Perish", pp. 337–338 [PDF pp. 356–357]. Reproduced verbatim (with the book's footnote 6). Answer: none in the book.

> **Publish or Perish:**[6] Imagine that any newly minted Ph.D. who starts a tenure-track assistant professor job (player 1) is one of two types: high-ability (θ_H) or low-ability (θ_L), where $\theta_H > \theta_L > 0$. The assistant professor knows his type, but the department that hires him (player 2) knows only that he has high ability with probability $p < \frac{1}{2}$. The assistant professor first chooses how hard to work, which is effectively how many papers to publish in period 1 (the pre-tenure period). After observing how many papers the assistant professor published, the department decides whether to grant him tenure (T) or not to do so (N). If the department chooses to grant tenure then the assistant professor's payoff is $v_1(q, T \mid \theta) = V - \frac{q}{\theta}$, where V is the value of being tenured. The department's payoff is 1 if it tenures a high-ability type and -1 if it tenures a low-ability type. If the department denies tenure, it gets a payoff of 0 and the assistant professor's payoff is $v_1(q, N \mid \theta) = -\frac{q}{\theta}$. Denote by $\mu(q)$ the department's belief that the professor is a high-ability type given that he published q papers.
>
> a. Write down a "rough" extensive form of this game.
> b. If there is a pooling perfect Bayesian equilibrium, will the assistant professor be tenured? Does he write any papers? What then is the unique outcome of the pooling perfect Bayesian equilibrium?
> c. Characterize the perfect Bayesian separating equilibria using the parameters of the game.
> d. If there is more than one separating perfect Bayesian equilibrium, what is the smallest number of papers that an assistant professor needs to publish in order to get tenure?
>
> [6] This exercise is based on one written by Rod Garrett and John Wooders.

(a) What is asked. (a) A rough extensive form; (b) whether a pooling PBE has tenure and papers, and its unique outcome; (c) the separating PBE in terms of the parameters; (d) the smallest number of papers that gets tenure.

(b) Method. A signaling game with a continuum of messages $q \geq 0$ (papers) and two replies T or N: the department tenures iff its belief $\mu(q) = \Pr(\theta_H \mid q)$ makes tenure pay, then incentive constraints for each type, with off-path beliefs chosen to deter deviations (Tadelis §16.1–16.2, pp. 319–332: the same logic as limit pricing). The department's payoff from T is $\mu - (1-\mu) = 2\mu - 1$, from N it is 0: it tenures iff $\mu(q) \geq \frac{1}{2}$.

(c) Solution. **(a) Extensive form (rough).** Nature picks θ_H (probability $p < \frac{1}{2}$) or θ_L; the assistant professor, who knows his type, picks $q \geq 0$; the department sees only q and picks T or N. Payoffs (professor, department): $(V - \frac{q}{\theta}, \pm 1)$ after T (with $+1$ for θ_H, -1 for θ_L) and $(-\frac{q}{\theta}, 0)$ after N. A different q is a different message, with its own information set containing both type nodes.

(b) Pooling. If both types publish the same $\bar{q}$, Bayes' rule gives $\mu(\bar{q}) = p < \frac{1}{2}$, so the department denies tenure: the professor is not tenured. Then the payoff is $-\bar{q}/\theta$, which a deviation to $q = 0$ improves to at least 0 (whatever belief follows $q = 0$ the payoff is 0 or V), so $\bar{q} = 0$. Conversely, $\bar{q} = 0$ is supported by off-path beliefs $\mu(q) < \frac{1}{2}$ for $q > 0$ (deviating to $q > 0$ yields $-q/\theta < 0$). The

unique outcome of a pooling PBE: nobody publishes, nobody is tenured, payoffs 0 for everybody; the department's belief p stays at the prior.

(c) Separating. θ_L publishes q_L, θ_H publishes $q_H \neq q_L$; Bayes' rule gives $\mu(q_H) = 1$ (tenure) and $\mu(q_L) = 0$ (no tenure).

- θ_L has no reason to publish: he is denied tenure, so $q_L = 0$ (otherwise he would gain by deviating to 0). (The reverse profile, where θ_H publishes less than θ_L, fails: θ_L would copy θ_H to be tenured at lower cost.)
- Write $q_H = q^* > 0$. *No mimicking by* θ_L*:* $0 \geq V - \frac{q^*}{\theta_L}$, i.e. $q^* \geq V\theta_L$. θ_H *is willing to publish:* $V - \frac{q^*}{\theta_H} \geq 0$, i.e. $q^* \leq V\theta_H$.
- Off the path take $\mu(q) = 0$ for $q < q^*$ and $\mu(q) = 1$ for $q > q^*$: fewer papers are not tenured and more papers only cost more. No type gains by moving to another q.

The separating PBE are exactly: $q_L = 0$, $q_H = q^* \in [V\theta_L,\ V\theta_H]$, tenure iff $q \geq q^*$ (or iff $q = q^*$), beliefs as above. The interval is nonempty because $\theta_H > \theta_L$: this is the single-crossing property, θ_H's cost per paper is lower.

(d) The smallest number of papers that gets tenure in a separating PBE is $q^* = V\theta_L$: θ_H publishes just enough to make θ_L indifferent about copying. All $q^* \in [V\theta_L, V\theta_H]$ are equilibria; $q^* = V\theta_L$ is the least-cost one.

(e) Verification. `check_T2.py` takes $V = 3$, $\theta_L = 1$, $\theta_H = 2$, $p = \frac{3}{10}$, a grid of papers $q \in \{0, \frac{1}{4}, \ldots, 10\}$ and checks every incentive constraint by brute force: the separating profiles are exactly $q^* \in \{3, \frac{13}{4}, \ldots, 6\}$ and the only pooling outcome is $q = 0$ with no tenure. Tadelis prints no answer.

Answer. (b) No: in a pooling PBE nobody is tenured and nobody publishes ($q = 0$, N). (c) $q_L = 0$, $q_H = q^* \in [V\theta_L, V\theta_H]$, tenure iff $q \geq q^*$, $\mu(q^*) = 1$, $\mu(0) = 0$. (d) $q^* = V\theta_L$.

Item 16: Tadelis Exercise 16.8

Exam · difficulty 4/5

Statement (verbatim from the list, item 16)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 16.8 "Beer or Quiche?", p. 338 [PDF p. 357]. Reproduced verbatim. Answer: none in the book.

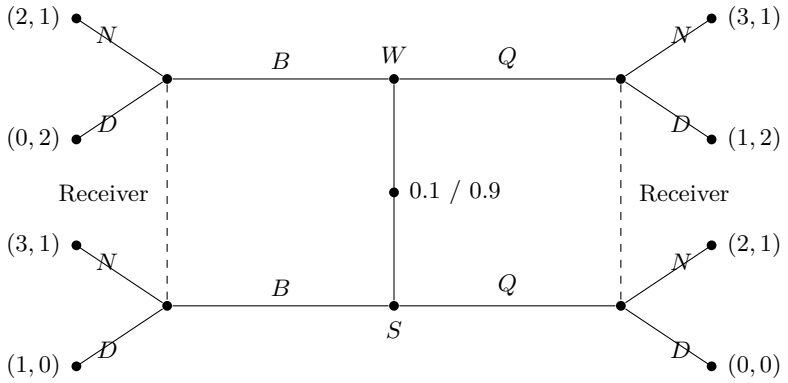
> **Beer or Quiche?** Cho and Kreps (1987) introduced what is now a famous two-player signaling game. First, Nature selects player 1, who knows his type, to be either a wimp (W), with probability $p = 0.1$, or surly (S), with probability $1 - p = 0.9$. Player 1 then chooses what to have for breakfast: beer (B) or quiche (Q). A surly type prefers beer while a wimp prefers quiche. Player 1's preferred breakfast gives him a payoff of 1 while his less-preferred choice gives him 0. After breakfast player 2 observes what player 1 ate but does not know whether he is a wimp or surly. Player 2 then chooses whether to duel (D) with player 1 or not to duel (N). Player 1, regardless of his type, prefers no duel, yielding him an extra payoff of 2, to a duel, which gives him 0. (For example, if player 1 eats his preferred breakfast and avoids a duel then his final payoff is 3, while if he eats his preferred breakfast and is forced into a duel then his final payoff is 1.) Player 2, however, prefers to duel if and only if player 1 is a wimp. If player 1 is surly then player 2's payoff is 0 from D and 1 from N. If player 1 is a wimp then player 2's payoff is 2 from D and 1 from N.
>
> a. Draw the extensive form of this game.
>
> b. What are the Bayesian Nash equilibria of this game?
>
> c. What are the perfect Bayesian equilibria?
>
> d. Of the equilibria you found in (c), which fail the intuitive criterion?

Note (errata, `.agent/notes/errata-game-theory-books.md`): Tadelis's Def. 16.1 (the intuitive criterion) prints v_2 in condition 2 where v_1 belongs; apply the corrected version (`Notas/L08/L08-08-intuitive-criterion.md`).

(a) What is asked. (a) The extensive form; (b) the BNE; (c) the PBE; (d) which PBE fail the intuitive criterion.

(b) Method. The Cho–Kreps game. Player 1 is Wimp (W, probability $p = \frac{1}{10}$) or Surly (S), picks Beer (B) or Quiche (Q); player 2 observes it and Duels (D) or not (N). Payoffs: breakfast 1 for the preferred one (S: B, W: Q), 0 for the other, plus 2 if no duel; player 2 gets 2 (D) or 1 (N) against W, and 0 (D) or 1 (N) against S. With $\mu = \Pr(W \mid \text{breakfast})$ player 2 gets 2μ from D and 1 from N: he duels iff $\mu \geq \frac{1}{2}$ (Tadelis §16.1 and §16.3, pp. 319–335; Gibbons §4.4, Signaling Requirement 6).

(c) Solution. **(a) Tree.** Payoffs (player 1, player 2):



(Left branches: message B; right branches: message Q; upper node W, lower node S; replies N above, D below. Upper-left $(2,1)$ is W with B and no duel: breakfast 0, no duel 2.)

Check of the payoffs: W with Q and N gets $1+2=3$, player 2 1; W with B and D gets 0, player 2 2; S with B and N gets 3, player 2 1; S with Q and D gets 0, player 2 0.

(b) BNE (pure). Strategies are (breakfast of W, breakfast of S) and (reply after B, reply after Q). The pure BNE are (BB; N after B, D after Q) and (QQ; D after B, N after Q). In the first, W gets 2 and S gets 3; a switch to Q would be met by a duel (payoffs 1 and 0). Symmetrically for the second, where W gets 3 and S gets 2. (Replies after the unreached breakfast may also be mixed, with $\Pr(D) \geq \frac{1}{2}$.)

(c) PBE: no separating, two pooling families. Let d_B, d_Q be the probabilities of a duel after B and Q. Then $u_S(B) - u_S(Q) = 1 - 2d_B + 2d_Q$ and $u_W(B) - u_W(Q) = -1 - 2d_B + 2d_Q$: *whatever the replies, Surly's gain from Beer exceeds Wimp's by* 2.

- *Separating* ($W\!:Q$, $S\!:B$): $\mu(Q) = 1$ so D, $\mu(B) = 0$ so N; W gets 1 from Q but 2 from B. Fails. *Reverse* ($W\!:B$, $S\!:Q$): D after B, N after Q; W gets 0 but 3 from Q. Fails.
- *Pooling on Beer.* Bayes' rule: $\mu(B) = p = 0.1 < \frac{1}{2}$, so N after B ($d_B = 0$). W gets 2 from B and $3 - 2d_Q$ from Q; he does not deviate iff $d_Q \geq \frac{1}{2}$, which requires $\mu(Q) \geq \frac{1}{2}$. S gets 3 from B and at most 2 from Q. PBE: both B; N after B; D after Q (or a duel with probability $d_Q \geq \frac{1}{2}$ at $\mu(Q) = \frac{1}{2}$); off-path belief $\mu(Q) = \Pr(W \mid Q) \geq \frac{1}{2}$.
- *Pooling on Quiche.* $\mu(Q) = 0.1$, so N after Q. S gets 2 from Q and $3 - 2d_B$ from B: he does not deviate iff $d_B \geq \frac{1}{2}$, so $\mu(B) = \Pr(W \mid B) \geq \frac{1}{2}$. W gets 3 from Q and at most 2 from B. PBE: both Q; N after Q; D after B; off-path belief $\mu(B) \geq \frac{1}{2}$.
- *Semi-separating.* If W mixes with positive probability on B, Surly strictly prefers B (the identity above), so $\mu(B) \leq p < \frac{1}{2}$ and $d_B = 0$; W's indifference then needs $d_Q = \frac{1}{2}$, but Q is chosen only by W ($\mu(Q) = 1$, $d_Q = 1$): contradiction. If W plays Q for sure and S mixes, $\mu(B) = 0$, $d_B = 0$ and Surly strictly prefers B: he does not mix.

These are all the PBE.

(d) Intuitive criterion. A message is equilibrium-dominated for a type if his equilibrium payoff exceeds the best he can get from it.

- *Pooling on Quiche fails.* W's equilibrium payoff is 3 and the best he can get from B is 2 (no duel): B is equilibrium-dominated for W. S's equilibrium payoff is 2 and B can give him 3: not dominated. So after B player 2 must believe Surly, $\mu(B) = 0$, and does not duel; then S gets $3 > 2$ from deviating to B. The equilibrium does not survive.
- *Pooling on Beer survives.* S's equilibrium payoff is 3 and Q gives him at most 2: Q is equilibrium-dominated for S. W gets 2 in equilibrium and up to 3 from Q: not dominated. So after Q player 2 must believe Wimp, $\mu(Q) = 1 \geq \frac{1}{2}$, and duels: W gets $1 < 2$. This is

the admissible belief the equilibrium needed, so no type deviates.

(The criterion is applied with Surly's and Wimp's payoffs, i.e. the sender's, as in Gibbons' Signaling Requirement 6; Tadelis prints v_2 where v_1 belongs in condition 2 of his Def. 16.1.)

(e) Verification. `check_T2.py` builds the matrix (pure BNE: the two above), enumerates all pure PBE on a grid of step $\frac{1}{60}$ (only the two pooling profiles, with off-path beliefs $\geq \frac{1}{2}$ on the unreached breakfast), checks the identity $u_S(B) - u_S(Q) = u_W(B) - u_W(Q) + 2$ for a grid of d_B, d_Q, and applies Requirement 6: only pooling on Quiche is eliminated. Tadelis prints no answer.

Answer. (b) Pure BNE: $(BB;\ N, D)$ and $(QQ;\ D, N)$ (replies after B, after Q). (c) PBE: pooling on Beer (N after B, D after Q, $\Pr(W \mid Q) \geq \frac{1}{2}$) and pooling on Quiche (N after Q, D after B, $\Pr(W \mid B) \geq \frac{1}{2}$); no separating PBE. (d) Pooling on Quiche fails the intuitive criterion; pooling on Beer survives.

Item 17: Gibbons Problem 4.16

Exam · difficulty 3/5

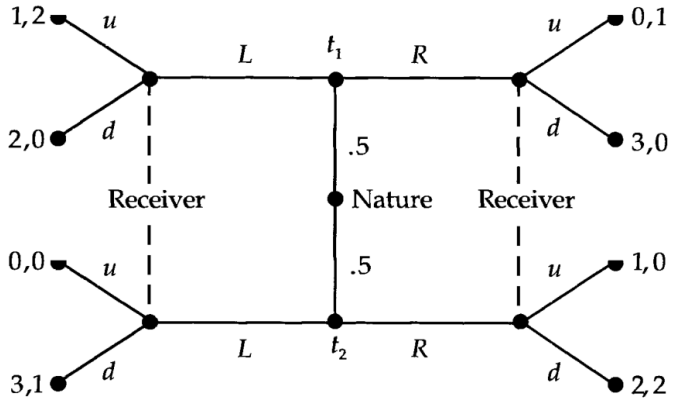
Statement (verbatim from the list, item 17)

Source. Gibbons (1992), *Game Theory for Applied Economists*, §4.4 Problems, Problem 4.16, p. 253 [PDF p. 259]. Reproduced verbatim. Problem 4.3, to which it refers, is copied below the statement (it is also item 5). Answer: none in the book.

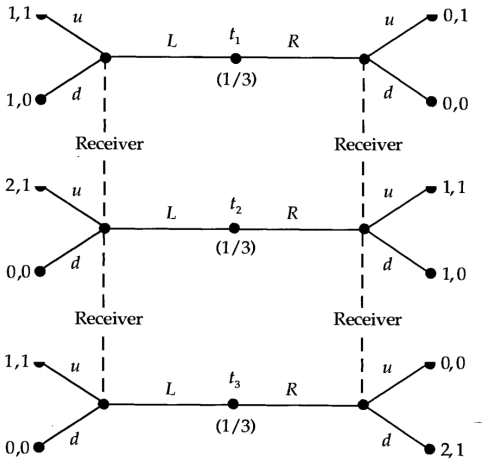
> 4.16. Consider the pooling equilibria described in Problem 4.3 (a) and (b). For each equilibrium: (i) determine whether the equilibrium can be supported by beliefs that satisfy Signaling Requirement 5; (ii) determine whether the equilibrium can be supported by beliefs that satisfy Signaling Requirement 6 (The Intuitive Criterion).

Problem 4.3 (§4.2 Problems, pp. 246–247 [PDF pp. 252–253]), copied from the book because the exercise refers to it.

> 4.3. a. Specify a pooling perfect Bayesian equilibrium in which both Sender types play R in the following signaling game.



b. The following three-type signaling game begins with a move by nature, not shown in the tree, that yields one of the three types with equal probability. Specify a pooling perfect Bayesian equilibrium in which all three Sender types play L.



(a) What is asked. For the pooling equilibria of Problem 4.3 (a) and (b): can each be supported by beliefs that satisfy Signaling Requirement 5, and by beliefs that satisfy Requirement 6 (the Intuitive Criterion)?

(b) Method. Gibbons §4.4, pp. 233–240. After an off-path message m_j, Requirement 5 puts zero probability on every type for which m_j is *dominated* (some other message gives that type a payoff whose lowest value exceeds the highest value from m_j); Requirement 6 does the same for every type for which m_j is *equilibrium-dominated* (the type's equilibrium payoff exceeds the highest payoff from m_j). The restriction applies unless it would exclude all types.

(c) Solution. **(a)** The equilibrium of item 5: both types play R; after R the Receiver plays d; equilibrium payoffs $U^*(t_1) = 3$, $U^*(t_2) = 2$; the off-path message is L, and the equilibrium needs the reply u after L, i.e. $\nu = \Pr(t_1 \mid L) \geq \frac{1}{3}$. Highest payoffs from L: t_1: $\max(1, 2) = 2$; t_2: $\max(0, 3) = 3$. Lowest payoffs from R: t_1: $\min(0, 3) = 0$; t_2: $\min(1, 2) = 1$.

- *Requirement 5.* L is dominated for t_1 iff $0 > 2$ (no) and for t_2 iff $1 > 3$ (no). Nothing is excluded, so the beliefs $\nu \geq \frac{1}{3}$ are admissible: **satisfied**.
- *Requirement 6.* L is equilibrium-dominated for t_1 ($3 > 2$) but not for t_2 ($2 < 3$). So $\nu = \Pr(t_1 \mid L) = 0$, hence $\nu < \frac{1}{3}$ and the Receiver plays d after L, which gives t_2 the payoff $3 > 2$: he deviates. **Violated**; the pooling equilibrium cannot be supported by beliefs that satisfy Requirement 6.

(b) The equilibrium: all three types play L; the Receiver plays u after L; $U^* = (1, 2, 1)$; the off-path message is R and the equilibrium needs u after R, i.e. $r_3 \leq \frac{1}{2}$ (the Receiver's payoff is $r_1 + r_2$ from u and r_3 from d). Highest payoffs from R: $(0, 1, 2)$ for t_1, t_2, t_3. Lowest payoffs from L: $(1, 0, 0)$.

- *Requirement 5.* R is dominated for t_1 ($1 > 0$) only. So $r_1 = 0$, and the beliefs $r = (0, r_2, r_3)$ with $r_3 \leq \frac{1}{2}$ (for instance $(0, \frac{2}{3}, \frac{1}{3})$) remain: **satisfied**.
- *Requirement 6.* R is equilibrium-dominated for t_1 ($1 > 0$) and for t_2 ($2 > 1$), not for t_3 ($1 < 2$). So $r = (0, 0, 1)$, $r_3 = 1 > \frac{1}{2}$, the Receiver plays d after R, and t_3 gets $2 > 1$ by deviating. **Violated**.

Both pooling equilibria satisfy Requirement 5 and violate Requirement 6: the extra power of the Intuitive Criterion comes from comparing with the *equilibrium* payoff, not with the worst payoff of another message.

(e) Verification. `check_T2.py` applies both requirements by enumeration (all PBE of the two games, beliefs on grids of steps $\frac{1}{60}$ and $\frac{1}{12}$): in each game the pooling profile survives the 'dominated' filter and is eliminated by the 'equilibrium-dominated' filter, and it recomputes the payoff comparisons used above. Gibbons prints no answer.

Answer. (a) Requirement 5: yes. Requirement 6: no (t_1 is excluded after L, the Receiver then plays d and t_2 deviates). (b) Requirement 5: yes (beliefs with $r_1 = 0$, $r_3 \leq \frac{1}{2}$). Requirement 6: no ($r = (0, 0, 1)$, the Receiver plays d, t_3 deviates).

Item 18: MWG Exercise 13.C.1

Advanced · difficulty 2/5

Statement (verbatim from the list, item 18)

Source. Mas-Colell, Whinston & Green (1995), *Microeconomic Theory*, Ch. 13 Exercises, Exercise 13.C.1^B, p. 474 [PDF p. 248]. Reproduced verbatim. Answer: none in the book; entries for 13.C.1, 13.C.2, 13.C.4 and 13.C.5 are in the MWG solutions manual (`Livros/`).

> **13.C.1^B** Consider a game in which, first, nature draws a worker's type from some continuous distribution on $[\underline{\theta}, \bar{\theta}]$. Once the worker observes her type, she can choose whether to submit to a costless test that reveals her ability perfectly. Finally, after observing whether the worker has taken the test and its outcome if she has, two firms bid for the worker's services. Prove that in any subgame perfect Nash equilibrium of this model all worker types submit to the test, and firms offer a wage no greater than $\underline{\theta}$ to any worker not doing so.

(a) What is asked. Prove that in every subgame perfect Nash equilibrium all worker types take the costless test, and that firms offer a wage no greater than $\underline{\theta}$ to a worker who does not.

(b) Method. Backward induction (unraveling): Bertrand competition for the worker at the last stage, then the worker's decision whether to be tested (MWG §13.C, pp. 450–451, which states the result; §13.B for the wage game). Types have an atomless distribution on $[\underline{\theta}, \bar{\theta}]$ with support the whole interval. A tested worker of type θ is offered θ: after the test the two firms know her productivity, and Bertrand competition drives the wage to it.

(c) Solution. Let N be the set of types that skip the test in the equilibrium and w the wage the two firms offer to a worker who skips it.

Step 1: $w \le E[\theta \mid N]$ whenever $\Pr(N) > 0$. Firms know only that the worker is in N. If the winning wage exceeded $E[\theta \mid N]$ the winning firm would lose money in expectation and would prefer to bid lower (to lose the worker); as in a Bertrand game with common values, the equilibrium wage equals $E[\theta \mid N]$, and in any case it is at most that.

Step 2: $N \subseteq [\underline{\theta}, w]$. A type who skips the test earns w; testing earns her θ. So every type in N has $\theta \le w$ (types above w test).

Step 3: $\Pr(N) = 0$. Suppose $\Pr(N) > 0$. By Step 2 all types in N are at most w, so $E[\theta \mid N] \le \sup N \le w$, and since the distribution has no atoms, N contains types strictly below $\sup N$ with positive conditional probability, so the first inequality is strict: $E[\theta \mid N] < w$. This contradicts Step 1, which gives $w \le E[\theta \mid N]$. So almost every type takes the test. (The intuition: whoever sits at the top of the non-testers earns less than her type from the average of those below, so she tests; then the next one does, and so on until nobody is left. For a uniform distribution the average of the types in $[a, x]$ is $\frac{a+x}{2} < x$.)

Step 4: the wage after no test. Suppose a firm offered $w > \underline{\theta}$ to untested workers. Then every type in $[\underline{\theta}, \min\{w, \bar{\theta}\})$ strictly prefers skipping the test to being tested ($w > \theta$); this set has positive probability, because the support is the whole interval, so $\Pr(N) > 0$, contradicting Step 3. Hence the wage offered after no test is at most $\underline{\theta}$: not taking the test is treated as being the worst type (only the lowest type may be indifferent between testing and not).

(e) Verification. `check_T3.py` verifies with sympy that $E[\theta \mid \ell \le \theta \le x] < x$ at $x = \frac{9}{10}$ and $\ell \in \{0, \frac{1}{5}\}$ for the uniform density and for $f(\theta) = 2\theta$ on $[0, 1]$, and that the uniform average

equals $\frac{a+x}{2}$. The MWG solutions manual's entry for 13.C.1 gives the same argument (the highest non-testing type prefers to test, so only the lowest type may not).

Answer. All types except possibly a null set (and $\underline{\theta}$, who is indifferent) take the test; firms offer an untested worker at most $\underline{\theta}$.

Item 19: MWG Exercise 13.C.5

Advanced · difficulty 2/5

Statement (verbatim from the list, item 19)

Source. Mas-Colell, Whinston & Green (1995), *Microeconomic Theory*, Ch. 13 Exercises, Exercise 13.C.5ˆB, p. 475 [PDF p. 248]. Reproduced verbatim. Answer: none in the book; entries for 13.C.1, 13.C.2, 13.C.4 and 13.C.5 are in the MWG solutions manual (`Livros/`).

> **13.C.5ˆB** Assume a single firm and a single consumer. The firm's product may be either high or low quality and is of high quality with probability λ. The consumer cannot observe quality before purchase and is risk neutral. The consumer's valuation of a high-quality product is v_H; her valuation of a low-quality product is v_L. The costs of production for high (H) and low (L) quality are c_H and c_L, respectively. The consumer desires at most one unit of the product. Finally, the firm's price is regulated and is set at p. Assume that $v_H > p > v_L > c_H > c_L$.
>
> (a) Given the level of p, under what conditions will the consumer buy the product?
>
> (b) Suppose that before the consumer decides whether to buy, the firm (which knows its type) can advertise. Advertising conveys no information directly, but consumers can observe the total amount of money that the firm is spending on advertising, denoted by A. Can there be a separating perfect Bayesian equilibrium, that is, an equilibrium in which the consumer rationally expects firms with different quality levels to pick different levels of advertising?

(a) What is asked. (a) When does the consumer buy at the regulated price p; (b) whether pure money-burning advertising A can support a separating PBE.

(b) Method. (a) Expected surplus of a risk-neutral consumer. (b) Single crossing: a signal separates only if it is more costly for the type that should not send it (MWG §13.C, pp. 455–457). Payoffs: the firm earns $p - c - A$ if the consumer buys and $-A$ if not; the consumer's surplus is $v - p$.

(c) Solution. **(a)** With probability λ the product is of high quality. The consumer buys iff $\lambda v_H + (1-\lambda)v_L \geq p$, i.e.

$$\lambda \;\geq\; \frac{p - v_L}{v_H - v_L} \in (0,1),$$

(the bound is in $(0,1)$ because $v_H > p > v_L$).

(b) Suppose a separating PBE exists: high quality advertises A_H, low quality $A_L \neq A_H$, and consumers infer the quality from A. After A_H the consumer believes high quality and buys ($v_H > p$); after A_L he believes low quality and does not buy ($v_L < p$). Then:

- the low-quality firm earns $-A_L$ in equilibrium (no sale) and the high-quality firm earns $p - c_H - A_H$, which must be at least what it gets by playing A_L instead (no sale: $-A_L$);
- the low-quality firm can imitate: spending A_H earns him $p - c_L - A_H$ because the consumer buys. Since $c_L < c_H$,

$$p - c_L - A_H \;>\; p - c_H - A_H \;\geq\; -A_L,$$

so imitating is strictly profitable.

The low-quality firm deviates, a contradiction: *no separating PBE exists*. Advertising cannot separate because it costs both types the same amount while the low-quality firm has the higher margin ($p - c_L > p - c_H$): the single-crossing property fails. Only pooling equilibria are possible (in the discrete example checked below there are 30 of them, all pooling).

(e) Verification. `check_T3.py` derives the threshold of (a) with sympy and, for $v_H = 10$, $p = 6$, $v_L = 4$, $c_H = 3$, $c_L = 1$, $\lambda = \frac{1}{2}$ and $A \in \{0, \ldots, 4\}$, enumerates all pure sender and receiver strategies with beliefs on a grid of step $\frac{1}{60}$: no separating PBE, while pooling profiles exist; it also checks the inequality $p - c_L - A > p - c_H - A$ for $A = 0, \ldots, 49$. The MWG solutions manual's entry for 13.C.5 gives the same argument.

Answer. (a) The consumer buys iff $\lambda \geq \frac{p - v_L}{v_H - v_L}$. (b) No: the low-quality firm profits from imitating the high-quality firm's advertising, since $c_L < c_H$.

Item 20: Tadelis Exercise 15.6

Advanced · difficulty 3/5

Statement (verbatim from the list, item 20)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 15.6 "Selten's Horse", p. 317 [PDF p. 336]. Reproduced verbatim. Answer: none in the book.

> **Selten's Horse:** Consider the three-person game described in Figure 15.6, known as Selten's horse (for the obvious reason).

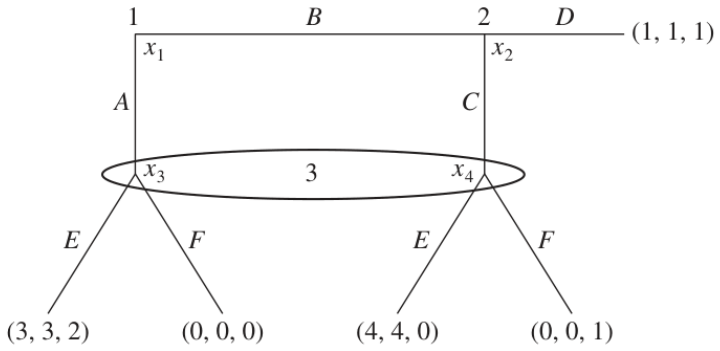


FIGURE 15.6 Selten's horse, exercise 15.6.

FIGURE 15.6 Selten's horse, exercise 15.6.

a. What are the pure-strategy Bayesian Nash equilibria of this game?

b. Which of the Bayesian Nash equilibria that you found in (a) are perfect Bayesian equilibria?

c. Which of the Bayesian Nash equilibria that you found in (a) are sequential equilibria?

(a) What is asked. (a) The pure-strategy BNE of Selten's horse; (b) which are PBE; (c) which are sequential equilibria.

(b) Method. List the eight pure profiles and test each for a profitable deviation; for PBE add a belief $\mu = \Pr(x_3)$ at player 3's information set and require sequential rationality at *every* information set, including player 2's off-path node; for sequential equilibrium require μ to be a limit of Bayes beliefs from totally mixed strategies (Tadelis §15.2–15.3, Defs. 15.4–15.5, pp. 303–313; MWG §9.C).

(c) Solution. Player 1 chooses A (to x_3) or B (to player 2 at x_2); player 2 chooses C (to x_4) or D (payoffs $(1,1,1)$); player 3, who cannot tell x_3 from x_4, chooses E or F. Payoffs: at x_3: E gives $(3,3,2)$, F gives $(0,0,0)$; at x_4: E gives $(4,4,0)$, F gives $(0,0,1)$.

(a) Pure BNE.

Profile	Payoffs	Profitable deviation?
(A, C, E)	$(3, 3, 2)$	player 1 to B: $(4, 4, 0)$ gives him $4 > 3$
(A, C, F)	$(0, 0, 0)$	player 3 at x_3: E gives $2 > 0$
(A, D, E)	$(3, 3, 2)$	none: B leads to D and gives player 1 $1 < 3$
(A, D, F)	$(0, 0, 0)$	player 3 at x_3: E gives $2 > 0$
(B, C, E)	$(4, 4, 0)$	player 3 at x_4: F gives $1 > 0$
(B, C, F)	$(0, 0, 1)$	player 2: D gives $1 > 0$
(B, D, E)	$(1, 1, 1)$	player 1 to A: E gives $3 > 1$
(B, D, F)	$(1, 1, 1)$	none: A then F gives player 1 $0 < 1$; player 2 gets $1 > 0$ from D

The pure BNE are (A, D, E) and (B, D, F). In both, player 2's choice at x_2 or player 3's at the other node is never tested.

(b) PBE. Player 3's payoffs are E: 2μ and F: $1 - \mu$, so E is optimal iff $\mu \geq \frac{1}{3}$.

- (A, D, E): x_3 is reached, so Bayes' rule gives $\mu = 1$ and E is optimal. But at x_2 (a singleton information set, off the path) player 2 compares D, which gives 1, with C, which leads to x_4 where player 3 plays E and gives player 2 4. D is not optimal given player 3's strategy: **not a PBE**. The equilibrium rests on the threat of D, which player 2 would not carry out.
- (B, D, F): neither x_3 nor x_4 is reached, so μ is free; F is optimal iff $\mu \leq \frac{1}{3}$. Player 2 gets 1 from D and 0 from C (followed by F): D is optimal; player 1 gets 1 from B and 0 from A (followed by F): optimal. **PBE** with $\mu \in [0, \frac{1}{3}]$.

(c) Sequential equilibrium. Perturb: player 1 plays A with probability ε_A and player 2 plays C with probability ε_C, both positive. The probability of x_3 is ε_A and of x_4 is $(1 - \varepsilon_A)\varepsilon_C$, so

$$\mu^k = \frac{\varepsilon_A}{\varepsilon_A + (1 - \varepsilon_A)\varepsilon_C}.$$

Letting the two trembles vanish at different rates gives any limit in $[0, 1]$ (for a target $\mu_0 > 0$, $\varepsilon_C = \varepsilon_A(1 - \mu_0)/\mu_0$ gives μ_0 in the limit). Hence every $\mu \leq \frac{1}{3}$ is consistent, and (B, D, F) with such a μ is a sequential equilibrium. In this game consistency removes nothing from the PBE: the equilibrium that fails (the one with D at x_2 against E) is removed already by sequential rationality at x_2, which a BNE ignores.

(e) Verification. `check_T3.py` computes the pure BNE by enumeration (exactly the two above), tests PBE conditions on a grid of beliefs of step $\frac{1}{60}$ (only (B, D, F), with $\mu \leq \frac{1}{3}$; at (A, D, E) player 2's best reply to E is C), and checks the tremble beliefs for the targets $0, \frac{1}{4}, \frac{1}{3}$. Tadelis prints no answer.

Answer. (a) (A, D, E) and (B, D, F). (b) Only (B, D, F) with $\mu \leq \frac{1}{3}$; (A, D, E) fails because D is not optimal for player 2 at x_2. (c) (B, D, F) with $\mu \leq \frac{1}{3}$ is also sequential, since any $\mu \in [0, 1]$ is a limit of trembles.

Item 21: Gibbons Problem 4.7

Advanced · difficulty 3/5

Statement (verbatim from the list, item 21)

Source. Gibbons (1992), *Game Theory for Applied Economists*, §4.2 Problems, Problem 4.7, p. 249 [PDF p. 255]. Reproduced verbatim. Answer: none in the book.

> 4.7. Draw indifference curves and production functions for a two-type job-market signaling model. Specify a hybrid perfect Bayesian equilibrium in which the high-ability worker randomizes.

(a) What is asked. Draw indifference curves and production functions for a two-type job-market signaling model, and specify a hybrid (semi-separating) PBE in which the high-ability worker randomizes.

(b) Method. The exercise gives no numbers, so a specification is chosen: Gibbons §4.2.B, pp. 190–205 (Spence's model with $y(\eta, e) = \eta$, education unproductive, and competing firms paying expected output). In a hybrid equilibrium the type that randomizes must be indifferent between its two messages; Bayes' rule at the shared message then fixes the mixing probability.

(c) Solution. Model. Ability $\eta \in \{1, 2\}$ (low $\eta_L = 1$, high $\eta_H = 2$) with $\Pr(\eta_H) = q = \frac{2}{3}$; output $y(\eta, e) = \eta$; education cost $c(\eta, e) = e^2/\eta$; wage $w(e)$ equal to expected output given the firms' belief $\mu(e) = \Pr(\eta_H \mid e)$: $w(e) = 1 + \mu(e)$. Worker utility $w - c(\eta, e)$.

Hybrid PBE. The low type plays $e = 0$. The high type plays $e^* = 1$ with probability $r = \frac{1}{2}$ and $e = 0$ with probability $\frac{1}{2}$.

- *Beliefs.* $\mu(1) = 1$, so $w(1) = 2$ (only the high type plays 1). At $e = 0$ Bayes' rule gives

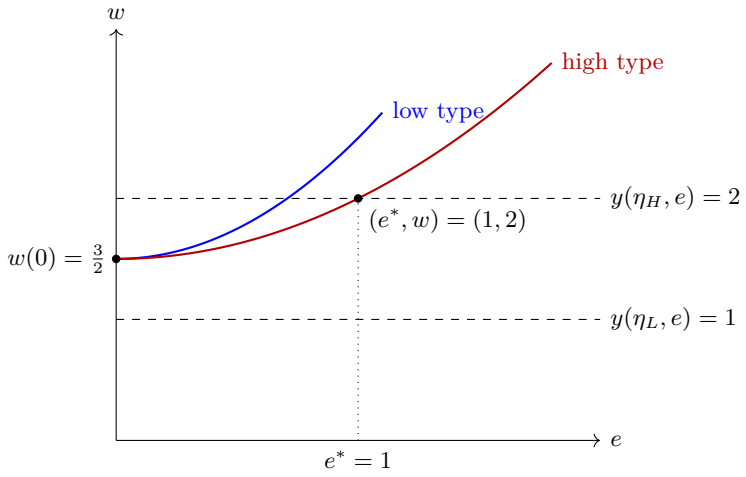
$$\mu(0) = \frac{q(1-r)}{q(1-r) + (1-q)} = \frac{\frac{2}{3} \cdot \frac{1}{2}}{\frac{2}{3} \cdot \frac{1}{2} + \frac{1}{3}} = \frac{1}{2}, \qquad w(0) = 1 + \frac{1}{2} = \frac{3}{2}.$$

  Off the path ($e \neq 0, 1$) take $\mu(e) = 0$, $w(e) = 1$.

- *High type indifferent:* $w(0) - c(2, 0) = \frac{3}{2}$ and $w(1) - c(2, 1) = 2 - \frac{1}{2} = \frac{3}{2}$.
- *Low type strictly prefers* 0: $\frac{3}{2}$ against $w(1) - c(1, 1) = 2 - 1 = 1$.
- *Deviations.* Any $e \in (0, 1)$ gets wage 1 and costs more than zero (payoffs $< \frac{3}{2}$ for both types); any $e > 1$ gets at most 2 even under the best belief, and $2 - \frac{e^2}{2} < \frac{3}{2}$, $2 - e^2 < \frac{3}{2}$.

Why this works: the mixing probability must make $\mu(0)$ low enough that the pooled wage $w(0)$ falls to $2 - c(2, e^*) = \frac{3}{2}$; $\mu(0) = 1 - \frac{e^{*2}}{2}$ requires $\mu(0) < q$ (the prior), which is why the high type has to play 0 with probability below one: $r > 0$.

Diagram. Wage against education. Production functions are the flat lines $w = \eta_L = 1$ and $w = \eta_H = 2$ (output does not depend on e). Indifference curves through the pooled point $(0, \frac{3}{2})$: $w = \frac{3}{2} + e^2$ for the low type and $w = \frac{3}{2} + \frac{e^2}{2}$ for the high type. The point $(1, 2)$ lies on the high type's curve and below the low type's curve ($2 < \frac{5}{2}$ at $e = 1$): this is single crossing.



(e) Verification. `check_T3.py` checks Bayes' rule exactly ($\mu(0) = \frac{1}{2}$, $w(0) = \frac{3}{2}$), the high type's indifference, the low type's strict preference and every deviation $e \in \{0, \frac{1}{20}, \ldots, 5\}$ under the off-path wage 1 (and under the best-case wage 2 for $e \geq 1$). Gibbons prints no answer.

Answer. Specification: $\eta \in \{1, 2\}$, $\Pr(\eta = 2) = \frac{2}{3}$, $y = \eta$, $c = e^2/\eta$. Hybrid PBE: low type $e = 0$; high type $e = 1$ with probability $\frac{1}{2}$ and $e = 0$ with probability $\frac{1}{2}$; $\mu(1) = 1$, $\mu(0) = \frac{1}{2}$, off-path $\mu = 0$; wages $w(1) = 2$, $w(0) = \frac{3}{2}$.

Item 22: Tadelis Exercise 16.7

Advanced · difficulty 4/5

Statement (verbatim from the list, item 22)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 16.7 "Intuitive Limit Pricing", p. 338 [PDF p. 357]. Reproduced verbatim. Answer: none in the book.

> **Intuitive Limit Pricing:** Show that of all the separating equilibria of the limit pricing–entry deterrence game in Section 16.2 only the best separating equilibrium does not fail the intuitive criterion.

Note: the game of Tadelis §16.2 (pp. 323–332) is worked in `Notas/L08/L08-07-limit-pricing.md` (inverse demand $5-q$, costs 1 and 2, separating $q_1^{1L} \in [2.62, 3.32]$). Errata: Tadelis p. 331 prints the entrant's pooled-entry profit as $\frac{7}{18}$, arithmetic gives $\frac{7}{36}$; Def. 16.1 condition 2 reads v_2 where v_1 belongs.

Game of Section 16.2 (pp. 323–324 [PDF pp. 342–343]), copied from the book because the exercise refers to it.

> Consider a market for some good that will last for two periods with demand $P = 5 - q$ in each period. In period $t = 1$ only one firm, the incumbent (player 1), is in the market, and this incumbent will continue to produce in period $t = 2$. A potential entrant (player 2) observes what happened in period $t = 1$ and then makes a choice of whether to enter the market at a fixed cost of $\frac{1}{2}$ and compete against the incumbent in period $t = 2$, or to remain out of the market. If player 2 stays out of the market then player 1 remains a monopolist in period 2, while if player 2 enters the market then they compete in a Cournot duopoly game. Each firm has marginal costs c_i, where it is common knowledge that the entrant's marginal costs are $c_2 = 2$. There is asymmetric information with respect to the costs (the type) of the incumbent. In particular the entrant knows only that $c_1 \in \{1, 2\}$, so that the incumbent can be either more efficient than the entrant (have low $[L]$ costs of $c_1 = 1$) or as efficient (have high $[H]$ costs of $c_1 = 2$). It is common knowledge that $\Pr\{c_1 = 1\} = \frac{1}{2}$, but only the incumbent knows his true costs.
>
> The timing of this game, which captures the story, is described as follows:
>
> 1. Nature chooses the type of player 1, $c_1 \in \{1, 2\}$, each type with equal probability.
> 2. Player 1 observes his costs and decides how much to produce in the first period, q_1^1, and the price in the first period is then $P = 5 - q_1^1$.
> 3. Player 2 observes q_1^1 and decides whether or not to enter at a fixed cost of $F = \frac{1}{2}$.
> 4. If player 2 stays out then in period $t = 2$ player 1 chooses q_1^2 and the price is $P = 5 - q_1^2$.
> 5. If player 2 enters then in period $t = 2$ each player i simultaneously chooses q_i^2 and the price is $P = 5 - q_1^2 - q_2^2$ (Cournot competition).

(a) What is asked. Show that, of all the separating equilibria of the limit-pricing game of §16.2, only the best separating equilibrium (the one with the lowest first-period output of the low-cost type) satisfies the intuitive criterion.

(b) Method. Tadelis §16.2 (pp. 323–332) for the separating set and §16.3 (pp. 332–335) for the criterion. The criterion, in the form of Cho–Kreps and Gibbons' Signaling Requirement 6: after an off-path output q', the entrant's belief must put zero weight on every type for which q' is equilibrium-dominated, unless that holds for both types. (Tadelis' Def. 16.1 prints v_2 in condition 2; the sender's payoff v_1 is meant.)

(c) Solution. The separating set. Costs 1 (L) and 2 (H), demand $5-q$, entry cost $\frac{1}{2}$, entrant's cost 2. Monopoly profits are 4 (L, at $q=2$) and $\frac{9}{4}$ (H, at $q=\frac{3}{2}$). In any separating PBE H plays its monopoly output $\frac{3}{2}$ and faces entry (Cournot profit 1): H's payoff is $\frac{9}{4}+1=\frac{13}{4}$. L produces q in period 1, deters entry, and then earns the monopoly 4: payoff $(4-q)q+4$. With the off-path belief "high cost" (entry follows any other output) the incentive constraints are

$$\text{H does not mimic: } (3-q)q+\tfrac{9}{4}\le\tfrac{13}{4} \iff q^2-3q+1\ge 0 \iff q\ge\tfrac{3+\sqrt{5}}{2}\approx 2.618,$$

$$\text{L does not deviate: } (4-q)q+4\ge\tfrac{25}{4} \iff q\le 2+\sqrt{\tfrac{7}{4}}\approx 3.323,$$

(L's best deviation is $q=2$, profit 4, followed by entry and the best reply to the entrant's 1, profit $\frac{9}{4}$: total $\frac{25}{4}$). The separating PBE are indexed by $q_1^{1L}=q\in[q_{\min},q_{\max}]=[2.618,\ 3.323]$, $q_{\min}=\frac{3+\sqrt{5}}{2}$, and the best one is $q_{\min}$.

Equilibrium-dominated outputs. Let q' be an off-path output. The most the entrant's reaction can give a type is no entry, so the highest payoffs are

$$\bar{u}_H(q')=(3-q')q'+\tfrac{9}{4},\qquad \bar{u}_L(q')=(4-q')q'+4.$$

q' is equilibrium-dominated for H iff $\bar{u}_H(q')<\frac{13}{4}$, i.e. $q'>q_{\min}$ (or $q'<0.382$); for L iff $\bar{u}_L(q')<(4-q)q+4$. Since $(4-x)x$ is symmetric about $x=2$, this holds iff $q'>q$ or $q'<4-q$: outputs between $4-q$ and q are not dominated for L.

Equilibria with $q>q_{\min}$ fail. Take $q'\in(q_{\min},q)$. It is equilibrium-dominated for H ($q'>q_{\min}$) but not for L: $\bar{u}_L(q')=(4-q')q'+4>(4-q)q+4$ because $2<q'<q$. The criterion forces the belief that the incumbent is low-cost, so the entrant stays out; then L earns $\bar{u}_L(q')>$ his equilibrium payoff and deviates. The equilibrium is not supported by admissible beliefs. This covers every q in $(q_{\min},q_{\max}]$.

The best equilibrium survives. At $q=q_{\min}$, any $q'>q_{\min}$ is equilibrium-dominated for H; and for L as well, since $\bar{u}_L(q')<(4-q_{\min})q_{\min}+4$ ($q'>q_{\min}>2$). Both types are dominated, so there is no restriction and the original off-path belief (high cost, entry) stays admissible. For $q'<q_{\min}$ that is equilibrium-dominated for H (only $q'<0.382$), L's best payoff $\bar{u}_L(q')\le 5.38$ is below its equilibrium payoff 7.618: again both are dominated. For $q'\in[0.382,q_{\min})$ H is not dominated, and the criterion requires nothing about L. In every case the belief "high cost" is admissible and no type gains from deviating. So $q_{\min}$ satisfies the criterion. Only $q_1^{1L}=\frac{3+\sqrt{5}}{2}\approx 2.62$ survives.

(e) Verification. `check_T3.py` solves both quadratics with sympy ($q_{\min}=2.6180$, $q_{\max}=3.3229$) and applies the criterion on a grid of step 0.001 over $q'\in[0,6]$ to 20 equally spaced separating equilibria in $(q_{\min},q_{\max}]$ and to $q_{\min}$ itself: all of the first group fail, $q_{\min}$ survives. Tadelis prints no answer; the note `Notas/L08/L08-07-limit-pricing.md` works the same game.

Answer. The separating PBE are $q_1^{1L}\in\left[\frac{3+\sqrt{5}}{2},\ 2+\sqrt{7/4}\right]\approx[2.62,3.32]$. For $q_1^{1L}>\frac{3+\sqrt{5}}{2}$ an output in between is equilibrium-dominated only for the high-cost type, so the entrant must believe low cost and stay out, and the low-cost type deviates. Only $q_1^{1L}=\frac{3+\sqrt{5}}{2}$ survives the intuitive criterion.

Item 23: Muñoz-Garcia Ch. 11, Exercise 4

Advanced · difficulty 4/5

Statement (verbatim from the list, item 23)

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, Ch. 11, Exercise 4 "Labor Market Signaling Game and Equilibrium Refinements", p. 500 [PDF p. 507]. Reproduced verbatim (statement only; the book's worked answer follows it). Answer: yes, the book's worked answer follows the exercise.

> Consider the following labor market signaling model. A worker with privately known productivity θ chooses an education level e. Upon observing e, each firm responds with a wage offer that corresponds to the worker's expected productivity (That is, we assume that the labor market is perfectly competitive, so firms pay the value of the expected marginal productivity to every worker. Otherwise, competing firms would have incentives to increase their salaries in order to attract workers with higher productivities.). The worker's payoff from receiving a wage w and acquiring an education level e, given an innate productivity θ, is $u(w, e \mid \theta) = w - \frac{e^3}{\theta}$, i.e., the cost of acquiring education, $\frac{e^3}{\theta}$, is convex, and decreases in the worker's innate productivity, θ (Note that, for simplicity, we assume that the worker's innate productivity, θ, is unaffected by the education level he acquires.).
>
> **Part (a)** Assuming that there are two equally likely types $\theta_1 = 1$ and $\theta_2 = 2$, characterize the set of all separating equilibria.
>
> **Part (b)** Which of these equilibria survive the Cho and Kreps' (1987) Intuitive Criterion?

(a) What is asked. (a) All separating equilibria of the labor-market signaling game with $u = w - e^3/\theta$, $\theta_1 = 1$, $\theta_2 = 2$ equally likely; (b) which of them survive the Cho–Kreps intuitive criterion.

(b) Method. Competitive wage $w(e) = E[\theta \mid e]$. In a separating PBE each type is paid its productivity; the incentive constraints bound the high type's education from below (the low type must not mimic) and above (the high type must want to separate); off-path beliefs "low type" make deviations as unattractive as possible. Then the criterion selects among them (Gibbons §4.2.B and §4.4; Tadelis §16.3; MWG §13.C).

(c) Solution. **(a) Separating PBE.** By Bayes' rule the firms pay $w = \theta_1 = 1$ after the low type's education e_1 and $w = \theta_2 = 2$ after the high type's $e_2 \neq e_1$. The low type chooses $e_1 = 0$: any $e_1 > 0$ costs without raising the wage, and the deviation to 0 pays at least $\theta_1 = 1$ whatever the belief. The high type chooses $e_2 = \bar{e} > 0$ and receives 2. Off the path take $w(e) = 1$ (the belief "low type") for $e \neq \bar{e}$.

- *Low type must not mimic:* $1 - 0 \geq 2 - \bar{e}^3/1 \iff \bar{e}^3 \geq 1 \iff \bar{e} \geq 1$.
- *High type must prefer to separate:* $2 - \bar{e}^3/2 \geq 1 - 0 \iff \bar{e}^3 \leq 2 \iff \bar{e} \leq 2^{1/3} \approx 1.26$. (His best deviation is $e = 0$, wage 1.)
- *No other deviations pay:* any $e \neq \bar{e}$ gets $w = 1$ and costs e^3/θ, so no type gains over $e = 0$ or $e = \bar{e}$.

The set of separating PBE is

$$\Big\{(e_1 = 0,\ e_2 = \bar{e}) : \bar{e} \in [1,\ 2^{1/3}]\Big\}, \quad w(0) = 1,\ w(\bar{e}) = 2,\ w(e) = 1 \text{ otherwise},$$

a continuum. Payoffs: 1 for the low type, $2 - \bar{e}^3/2$ for the high type (between $2 - \frac{1}{2} = \frac{3}{2}$ and 1).

(b) Intuitive criterion. The equilibrium payoffs are $U^*(\theta_1) = 1$ and $U^*(\theta_2) = 2 - \bar{e}^3/2$. The highest wage any belief can give is 2, so the highest payoff from an off-path e is $2 - e^3$ for θ_1 and $2 - e^3/2$ for θ_2.

- e is equilibrium-dominated for θ_1 iff $1 > 2 - e^3$, i.e. $e > 1$; for θ_2 iff $2 - \bar{e}^3/2 > 2 - e^3/2$, i.e. $e > \bar{e}$.
- Suppose $\bar{e} > 1$ and take $e \in (1, \bar{e})$. It is dominated for θ_1 ($e > 1$) but not for θ_2 ($e < \bar{e}$). The criterion forces the belief θ_2, wage 2, and then θ_2 gets $2 - e^3/2 > 2 - \bar{e}^3/2$: he deviates. So every $\bar{e} > 1$ fails.
- $\bar{e} = 1$: for $e > 1$ the message is dominated for both types (no restriction); for $e \leq 1$ it is not dominated for θ_1 ($2 - e^3 \geq 1$), so there is no restriction either. The off-path belief "low type" stays admissible and nobody deviates: it survives.

Only the least-cost separating equilibrium $(e_1, e_2) = (0, 1)$, with wages $(1, 2)$, survives the intuitive criterion (high type's payoff $\frac{3}{2}$).

(e) Verification. `check_T3.py` solves the two cubic inequalities with sympy and checks, on a grid of step 0.01 for $\bar{e} \in [0, 2]$ and 0.002 for deviations, that the separating PBE are exactly $\bar{e} \in [1, 2^{1/3}]$, and then applies the criterion on a grid of step 0.001: $\bar{e} = 1$ survives, $\bar{e} \in \{1.01, 1.1, 1.2, 2^{1/3}\}$ fail. The book's printed answer (pp. 500–505) gets the same set and the same selection of the lowest level.

Answer. (a) $e_1 = 0$, $e_2 \in [1, 2^{1/3}]$ (wages 1 and 2; off-path belief "low type"). (b) Only $e_2 = 1$ survives the intuitive criterion.

Item 24: MWG Exercise 13.C.2

Advanced · difficulty 4/5

Statement (verbatim from the list, item 24)

Source. Mas-Colell, Whinston & Green (1995), *Microeconomic Theory*, Ch. 13 Exercises, Exercise 13.C.2ˆC, p. 475 [PDF p. 248]. Reproduced verbatim. Answer: none in the book; entries for 13.C.1, 13.C.2, 13.C.4 and 13.C.5 are in the MWG solutions manual (`Livros/`).

> **13.C.2ˆC** Reconsider the two-type signaling model with $r(\theta_L) = r(\theta_H) = 0$, assuming a worker's productivity is $\theta(1 + \mu e)$ with $\mu > 0$. Identify the separating and pooling perfect Bayesian equilibria, and relate them to the perfect information competitive outcome.

Model of Section 13.C (pp. 450–451 [PDF p. 236]), copied from the book because the exercise refers to it.

> Consider the following adaptation of the model discussed in Section 13.B. For simplicity, we restrict attention to the case of two types of workers with productivities θ_H and θ_L, where $\theta_H > \theta_L > 0$ and $\lambda = \text{Prob}\,(\theta = \theta_H) \in (0, 1)$. The important extension of our previous model is that before entering the job market a worker can get some education, and the amount of education that a worker receives is observable. To make matters particularly stark, we assume that education does *nothing* for a worker's productivity (see Exercise 13.C.2 for the case of productive signaling). The cost of obtaining education level e for a type θ worker (the cost may be of either monetary or psychic origin) is given by the twice continuously differentiable function $c(e, \theta)$, with $c(0, \theta) = 0$, $c_e(e, \theta) > 0$, $c_{ee}(e, \theta) > 0$, $c_\theta(e, \theta) < 0$ for all $e > 0$, and $c_{e\theta}(e, \theta) < 0$ (subscripts denote partial derivatives). Thus, both the cost and the marginal cost of education are assumed to be lower for high-ability workers; for example, the work required to obtain a degree might be easier for a high-ability individual. Letting $u(w, e \mid \theta)$ denote the utility of a type θ worker who chooses education level e and receives wage w, we take $u(w, e \mid \theta)$ to equal her wage less any educational costs incurred: $u(w, e \mid \theta) = w - c(e, \theta)$. As in Section 13.B, a worker of type θ can earn $r(\theta)$ by working at home.
>
> A portion of the game tree for this model is shown in Figure 13.C.1. Initially, a random move of nature determines whether a worker is of high or low ability. Then, conditional on her type, the worker chooses how much education to obtain. After obtaining her chosen education level, the worker enters the job market. Conditional on the observed education level of the worker, two firms simultaneously make wage offers to her. Finally, the worker decides whether to work for a firm and, if so, which one.

(a) What is asked. In the two-type signaling model with $r(\theta_L) = r(\theta_H) = 0$ and productivity $\theta(1 + \mu e)$, $\mu > 0$: identify the separating and pooling PBE and relate them to the perfect-information competitive outcome.

(b) Method. Same steps as MWG §13.C (pp. 450–459): firms pay expected productivity, so the wage after e is $E[\theta \mid e]\,(1 + \mu e)$, between $\theta_L(1 + \mu e)$ and $\theta_H(1 + \mu e)$. The worst off-path belief (low type) gives the wage $\theta_L(1 + \mu e)$. Incentive constraints for each type as in the unproductive case, but now education also raises the wage.

(c) Solution. Perfect-information outcome. If types were observed, type θ would choose $e^*(\theta)$ to maximize $\theta(1 + \mu e) - c(e, \theta)$, i.e. $\theta\mu = c_e(e^*, \theta)$, and earn the wage $w^*(\theta) = \theta(1 + \mu e^*(\theta))$; utility $u^*(\theta)$. This is the efficient level of education, and $e^*(\theta_H) > e^*(\theta_L)$ by $c_{e\theta} < 0$.

Separating PBE. By Bayes' rule each type is paid its productivity. The low type must choose $e_L = e^*(\theta_L)$: after e_L his wage is $\theta_L(1+\mu e_L)$, and any e off the path gets at least $\theta_L(1+\mu e)$, so he can always secure $u^*(\theta_L)$ and nothing more. The high type chooses e_H with the wage $\theta_H(1+\mu e_H)$, subject to

$$\begin{aligned}(\text{IC}_L) \quad & \theta_H(1+\mu e_H) - c(e_H,\theta_L) \le u^*(\theta_L),\\ (\text{IC}_H) \quad & \theta_H(1+\mu e_H) - c(e_H,\theta_H) \;\ge\; \max_e \,[\theta_L(1+\mu e) - c(e,\theta_H)],\end{aligned}$$

the right-hand side of (IC_H) being his best deviation under the belief "low type". By single crossing the set is an interval $e_H \in [\underline{e},\bar{e}]$, with off-path belief "low type". *Relation to the perfect-information outcome:* the pair $(e^*(\theta_L), e^*(\theta_H))$ is itself a separating PBE iff (IC_L) holds at $e_H = e^*(\theta_H)$, that is, iff the low type does not want to mimic the efficient high-type education (the constraint (IC_H) always holds there, because $e^*(\theta_H)$ maximizes his utility under his own wage). When (IC_L) fails at $e^*(\theta_H)$, then $\underline{e} > e^*(\theta_H)$: the high type *over-invests* in education to separate, and the outcome is below the perfect-information one. The low type always gets his perfect-information outcome.

Pooling PBE. Both types choose e and are paid $\bar{\theta}(1+\mu e)$ with $\bar{\theta} = \lambda\theta_H + (1-\lambda)\theta_L$. Deviations are met by the belief "low type", so each type compares with his best deviation: the low type's is $u^*(\theta_L)$, the high type's is the right-hand side of (IC_H). The pooling set is an interval $e \in [e_{\min}, e_{\max}]$ (including 0 when $\bar{\theta} \ge$ both deviation payoffs); the low type's constraint binds first. Pooling is inefficient: both types get the same education, and the competitive outcome with perfect information is not attained.

A worked example. $c(e,\theta) = e^2/\theta$, $\theta_L = 1$, $\theta_H = 2$, $\mu = \frac{1}{2}$, $\lambda = \frac{1}{2}$. Then $e^*(\theta) = \theta^2\mu/2$: $e^*(\theta_L) = \frac{1}{4}$, $e^*(\theta_H) = 1$, with $u^*(\theta_L) = \frac{17}{16}$ and $u^*(\theta_H) = \frac{5}{2}$.

- (IC_L): $2 + e - e^2 \le \frac{17}{16}$, i.e. $e \ge \underline{e} = \frac{2+\sqrt{19}}{4} \approx 1.59 > 1 = e^*(\theta_H)$: the full-information outcome is *not* sustainable, because the low type would gain by mimicking education 1 ($2 + 1 - 1 = 2 > \frac{17}{16}$).
- (IC_H): the best deviation under the low-type belief is $\frac{9}{8}$ (at $e = \frac{1}{2}$), and $2 + e - \frac{e^2}{2} \ge \frac{9}{8}$ gives $e \le \bar{e} = 1 + \frac{\sqrt{11}}{2} \approx 2.66$.
- Separating PBE: $e_L = \frac{1}{4}$ and $e_H \in [1.59,\ 2.66]$.
- Pooling: with $\bar{\theta} = \frac{3}{2}$ the low type needs $\frac{3}{2} + \frac{3}{4}e - e^2 \ge \frac{17}{16}$, i.e. $e \le \frac{3+\sqrt{37}}{8} \approx 1.135$; the high type needs $e \le \frac{3+\sqrt{21}}{4} \approx 1.90$. So pooling PBE exist for $e \in [0, 1.135]$.

The full-information outcome is attainable as a separating equilibrium for other parameters: for $\theta_H = 4$ (all else equal) $e^*(\theta_H) = 4$ and the low type's mimic payoff is $12 - 16 = -4 \le \frac{17}{16}$, so $(e^*(\theta_L), e^*(\theta_H))$ is itself a separating PBE (the case of Figure 13.C.2(a) of the solutions manual).

(e) Verification. `check_T3.py` recomputes e^*, u^*, the best deviation $\frac{9}{8}$, the two separating bounds and the two pooling bounds symbolically with sympy for the example, and checks the $\theta_H = 4$ case. The MWG solutions manual gives the same structure (separating (e_L^*, e_H) with e_H in an interval $[e', e'']$, pooling below a bound, the perfect-information outcome sustained only in some cases).

Answer. Perfect information: $\theta\mu = c_e(e^*,\theta)$, wage $\theta(1+\mu e^*)$. Separating PBE: low type at $e^*(\theta_L)$, high type at any $e_H \in [\underline{e},\bar{e}]$ defined by (IC_L) and (IC_H); it attains the perfect-information outcome iff the low type does not mimic $e^*(\theta_H)$, otherwise the high type over-invests. Pooling PBE: $e \in [e_{\min}, e_{\max}]$, inefficient. Example above: separating $e_H \in [1.59, 2.66]$, pooling $e \in [0, 1.135]$.

Item 25: MWG Exercise 13.C.4

Advanced · difficulty 4/5

Statement (verbatim from the list, item 25)

Source. Mas-Colell, Whinston & Green (1995), *Microeconomic Theory*, Ch. 13 Exercises, Exercise 13.C.4ˆB, p. 475 [PDF p. 248]. Reproduced verbatim. The upper bound's accent is blurred in the scan; read as $\bar{\theta}$, the notation of §13.C and of 13.C.1. Answer: none in the book; entries for 13.C.1, 13.C.2, 13.C.4 and 13.C.5 are in the MWG solutions manual (`Livros/`).

> **13.C.4ˆB** Reconsider the signaling model discussed in Section 13.C, now assuming that worker types are drawn from the interval $[\underline{\theta}, \bar{\theta}]$ with a density function $f(\theta)$ that is strictly positive everywhere on this interval. Let the cost function be $c(e, \theta) = (e^2/\theta)$. Derive the (unique) perfect Bayesian equilibrium.

Model of Section 13.C (pp. 450–451 [PDF p. 236]), copied from the book because the exercise refers to it.

> Consider the following adaptation of the model discussed in Section 13.B. For simplicity, we restrict attention to the case of two types of workers with productivities θ_H and θ_L, where $\theta_H > \theta_L > 0$ and $\lambda = \text{Prob}\,(\theta = \theta_H) \in (0, 1)$. The important extension of our previous model is that before entering the job market a worker can get some education, and the amount of education that a worker receives is observable. To make matters particularly stark, we assume that education does *nothing* for a worker's productivity (see Exercise 13.C.2 for the case of productive signaling). The cost of obtaining education level e for a type θ worker (the cost may be of either monetary or psychic origin) is given by the twice continuously differentiable function $c(e, \theta)$, with $c(0, \theta) = 0$, $c_e(e, \theta) > 0$, $c_{ee}(e, \theta) > 0$, $c_\theta(e, \theta) < 0$ for all $e > 0$, and $c_{e\theta}(e, \theta) < 0$ (subscripts denote partial derivatives). Thus, both the cost and the marginal cost of education are assumed to be lower for high-ability workers; for example, the work required to obtain a degree might be easier for a high-ability individual. Letting $u(w, e \mid \theta)$ denote the utility of a type θ worker who chooses education level e and receives wage w, we take $u(w, e \mid \theta)$ to equal her wage less any educational costs incurred: $u(w, e \mid \theta) = w - c(e, \theta)$. As in Section 13.B, a worker of type θ can earn $r(\theta)$ by working at home.
>
> A portion of the game tree for this model is shown in Figure 13.C.1. Initially, a random move of nature determines whether a worker is of high or low ability. Then, conditional on her type, the worker chooses how much education to obtain. After obtaining her chosen education level, the worker enters the job market. Conditional on the observed education level of the worker, two firms simultaneously make wage offers to her. Finally, the worker decides whether to work for a firm and, if so, which one.

Optional extras.

- Tadelis 15.4 "Campaigning" (p. 316): a second two-type signaling game with commitment by the rival, after item 10.

- Tadelis 16.1 (p. 335): uniqueness of beliefs in separating PBE of the item 10 game.

- Tadelis 16.2 "Limit Pricing Revisited" (p. 335): separation at monopoly quantities with a discount factor δ. As printed, the second quantity reads $q_1^{1L} = 1.5$; by Claim 16.2 it is the high-cost type's, $q_1^{1H} = 1.5$.

- Gibbons 4.5 (all pure PBE of Problem 4.3) and 4.6 (a signaling version of Fig. 4.1.1), p. 249.

- Muñoz-Garcia Ch. 9 Ex. 3–4 (pp. 394, 406: Spence with productivity- and utility-enhancing education), Ex. 6 (p. 432: cost uncertainty), Ex. 7 (p. 437: three types, three messages); Ch. 11 Ex. 1–3 (pp. 482, 487, 497) and Ex. 6 (p. 510). Pages from the PDF bookmarks with the registry offsets; not opened.

(a) What is asked. With types distributed on $[\underline{\theta}, \bar{\theta}]$ with a strictly positive density and cost $c(e, \theta) = e^2/\theta$, derive the (unique) separating PBE.

(b) Method. A continuum of types: in a separating PBE the wage schedule $w(e)$ must make each type's best choice reveal it, which gives a differential equation from the first-order condition plus the zero-profit condition $w(e(\theta)) = \theta$ (MWG §13.C, pp. 454–459 for the two-type version). The boundary condition comes from the lowest type. (The statement as printed says "perfect Bayesian equilibrium"; the MWG solutions manual notes in its first-printing errata that "separating" is meant.)

(c) Solution. Let $w(\cdot)$ be the firms' wage schedule and $e(\theta)$ the education of type θ, taken differentiable and increasing as in the text (separation requires different types to choose different e).

1. *First-order condition.* Type θ maximizes $w(e) - e^2/\theta$, so $w'(e(\theta)) = \dfrac{2e(\theta)}{\theta}$.
2. *Zero profit.* Competition implies $w(e(\theta)) = \theta$ (a revealed type is paid its productivity). Substituting $\theta = w(e)$ in the first-order condition:

$$w(e)\, w'(e) = 2e \iff \frac{d}{de}\, w(e)^2 = 4e \iff w(e)^2 = 2e^2 + k.$$

3. *Boundary condition.* The lowest type is paid $\underline{\theta}$ whatever he does (any belief gives at least $\underline{\theta}$), so he chooses $e(\underline{\theta}) = 0$ and $w(0) = \underline{\theta}$, so $k = \underline{\theta}^2$.

$$\boxed{w(e) = \sqrt{2e^2 + \underline{\theta}^2}, \qquad e(\theta) = \sqrt{\frac{\theta^2 - \underline{\theta}^2}{2}}}$$

with zero profit $w(e(\theta)) = \theta$. Sufficiency: type θ's utility $w(e') - e'^2/\theta$ has derivative $2e'(1/w(e') - 1/\theta)$, positive for $w(e') < \theta$, i.e. $e' < e(\theta)$, and negative afterwards: $e(\theta)$ is the global maximizer. Beliefs: $\mu(e) =$ the type with $e(\theta) = e$ (for e in the range) and, off the range ($e > e(\bar{\theta})$), the belief $\bar{\theta}$ (wage $\bar{\theta}$), which no one wants. Uniqueness: the ODE $ww' = 2e$ has the one-parameter family of solutions above and the boundary condition fixes it (the lowest type cannot choose $e > 0$, otherwise he would deviate to 0).

Compare with the manual. The MWG solutions manual writes $w(e) = \sqrt{2}\, e$ and $e(\theta) = \theta/\sqrt{2}$. That is the solution with $k = 0$, which satisfies $w(0) = 0$; it coincides with the formula above only if $\underline{\theta} = 0$ (it satisfies the differential equation and the first-order condition, but not the lowest type's boundary condition $w(0) = \underline{\theta} > 0$). With $\underline{\theta} > 0$ the formula above is the one that satisfies all conditions.

(e) Verification. `check_T3.py` solves the ODE with sympy (the family $w^2 = 2e^2 + C_1$), checks that the boxed w satisfies $ww' = 2e$, $w(0) = \underline{\theta}$, $w(e(\theta)) = \theta$, $e(\underline{\theta}) = 0$, that $e(\theta)$ solves the first-order condition, that it is the global maximizer for three $(\underline{\theta}, \theta)$ pairs on a grid of step 0.005, and that $w = \sqrt{2}e$ is the case $\underline{\theta} = 0$.

Answer. $w(e) = \sqrt{2e^2 + \underline{\theta}^2}$ and $e(\theta) = \sqrt{(\theta^2 - \underline{\theta}^2)/2}$; the lowest type chooses $e = 0$ and the education level rises with ability. (The manual's $w = \sqrt{2}e$, $e = \theta/\sqrt{2}$ is the case $\underline{\theta} = 0$.)